

Fieldbus Parameter Reference Dictionary

FB09-611A

R611

02/01

Notices and Trademarks

**Copyright 2000 by Honeywell International Inc.
Release R611 September 07, 2000**

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell and **TotalPlant** are registered trademarks of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell
Industrial Automation and Control
16404 North Black Canyon Highway
Phoenix, AZ 85053
1-800 343-0228

About This Document

This document supplies a listing of and cross reference to all of the Fieldbus parameters.

Release Information

Document Name	Document ID	Release Number	Publication Date
Parameter Reference Dictionary - FB09	FB09-611A	R611	02/01

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
PM Family Parameter Reference Dictionary	XPM09-611
TPS Fieldbus Theory	FB90-610
TPS Fieldbus Reference	FB40-610
TPS Fieldbus Overview	FB27-610
TPS Fieldbus Fundamentals	FB01-610

Contacts

World Wide Web

The following lists Honeywell's World Wide Web sites that will be of interest to our industrial automation and control customers.

Honeywell Organization	WWW Address (URL)
Corporate	http://www.honeywell.com
Industrial Automation and Control	http://www.iac.honeywell.com
International	http://content.honeywell.com/global/

Telephone

Contact us by telephone at the numbers listed below.

	Organization	Phone Number	
United States and Canada	Honeywell Inc. Industrial Automation and Control	1-800-343-0228	Sales Service Tech. Support
		1-800-525-7439	
		1-800-423-9883	
Asia Pacific	Honeywell Asia Pacific Inc. Hong Kong	(852) 28298298	
Europe	Honeywell PACE Brussels, Belgium	[32-2] 728-2111	
Latin America	Honeywell Inc. Sunrise, Florida U.S.A.	(954) 845-2600	

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

Symbol	Definition
	WARNING: Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

Contents

FIELDBUS PARAMETER CROSS-REFERENCES.....	9
INTRODUCTION.....	47
FIELDBUS PARAMETER DEFINITIONS (“A” THROUGH “Y”).....	59
A.....	59
B.....	66
C.....	75
D.....	87
E.....	97
F.....	100
G.....	111
H.....	113
I.....	118
J.....	128
L.....	129
M.....	137
N.....	143
O.....	146
P.....	155

Q 165

R..... 166

S..... 174

T 194

U..... 201

V..... 202

W..... 205

X..... 207

Y..... 211

APPENDIX A —ENUMERATIONS..... 213

1. Fieldbus Parameter Cross-references

1.1 Overview

This section contains two tables. The first table shows an alphabetical list of TPS Parameters in the left column with the corresponding FF Parameter Source parameters in the right column. This table, after several pages, is followed by one that alphabetically lists the FF Parameter Source first, which makes it easier to find a given FF Parameter Source.

1.2 TPS/Fieldbus Cross-Reference

The following table provides a TPS to fieldbus cross-reference listing.

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
\$AUXUNIT	None
\$OPTOL	None
\$SPTOL	None
ACTSCHVR	MIB: Network Management
ACTVSCHD	NMIB: LinkScheduleActivationVariable
ALENBST	None
ALMINFO	None
ALPRIOR	None
APPL	None
ASSOCCM	None
ASSOCDSP	None
BADPVFL	AI: OUT.Status
BADPVPR	None
BLKDDID	None
BLKODIDX	Block OD index
CASIN	CAS_IN
CASIND	CAS_IN_D
CONTCUT	None
CVEUHI	PID: OUT_SCALE.EU100
CVEULO	PID: OUT_SCALE.EU0
DEVHIFL	ALARM_SUM.Current
DEVHIPR	DV_HI_PRI
DEVLOFL	ALARM_SUM.Current

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
DEVELOPR	DV_LO_PRI
DEVSTAT	None.
DISPTYPE	None
EUDESC	None
FBTYPE	None
FFALMCUR	ALARM_SUM.Current
FFBCINST	BKCAL_IN.Status
FFBCINVL	BKCAL_IN.Value
FFBCOPST	BKCAL_OUT.Status
FFBCOPVL	BKCAL_OUT.Value
FFCSIDST	CAS_IN_D.Status
FFCSIDVL	CAS_IN_D.Value
FFCSINST	CAS_IN.Status
FFCSINVL	CAS_IN.Value
FFDEVREV	VFD revision number
FFDVHILM	DV_HI_LIM
FFDVHIPR	DV_HI_PRI
FFDVLOLM	DV_LO_LIM
FFDVLOPR	DV_LO_PRI
FFFFDECP	FF_SCALE.DecPt
FFFFGAIN	FF_GAIN
FFFFS0	FF_SCALE.EU0
FFFFS100	FF_SCALE.EU100
FFFFUIDX	FF_SCALE.UnitIndex
FFFFVLST	FF_VAL.Status
FFFFVLVL	FF_VAL.Value
FFGAIN	GAIN
FFHHLM	HI_HI_LIM
FFHHPR	HI_HI_PRI
FFHILM	HI_LIM
FFHIPR	HI_PRI

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
FFINSTS	IN.Status
FFINVAL	IN.Value
FFLLLM	LO_LO_LIM
FFLLPR	LO_LO_PRI
FFLOLM	LO_LIM
FFLOPR	LO_PRI
FFMODEA	MODE_BLK.Actual
FFMODEN	MODE_BLK.Normal
FFMODEP	MODE_BLK.Permitted
FFMODET	MODE_BLK.Target
FFOEXPST	OUT_EXP.Status
FFOEXPVL	OUT_EXP.Value
FFOPDECP	OUT_SCALE.DecPt
FFOPHILM	OUT_HI_LIM
FFOPLOLM	OUT_LO_LIM
FFOPS0	OUT_SCALE.EU0
FFOPS100	OUT_SCALE.EU100
FFOPUIDX	OUT_SCALE.UnitIndex
FFOREMST	OUT_REM.Status
FFOREMVL	OUT_REM.Value
FFOUT1ST	OUT_1.Status
FFOUT1VL	OUT_1.Value
FFOUT2ST	OUT_2.Status
FFOUT2VL	OUT_2.Value
FFOUTAST	OUT_ACCUM.Status
FFOUTAVL	OUT_ACCUM.Value
FFOUTDST	OUT_D.Status
FFOUTDVL	OUT_D.Value
FFOUTSTS	OUT.Status
FFOUTVAL	OUT.Value
FFPTRPST	OUT_PTRIP.Status

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
FFPTRPVL	OUT_PTRIP.Value
FFPVDECP	PV_SCALE.DecPt
FFPVS0	PV_SCALE.EU0
FFPVS100	PV_SCALE.EU100
FFPVSTS	PV.Status
FFPVUIDX	PV_SCALE.UnitIndex
FFPVVAL	PV.Value
FFRATE	RATE
FFRBDST	READBACK_D.Status
FFRBDVL	READBACK_D.Value
FFRBST	READBACK.Status
FFRBVL	READBACK.Value
FFRCIDST	RCAS_IN_D.Status
FFRCIDVL	RCAS_IN_D.Value
FFRCINST	RCAS_IN.Status
FFRCINVL	RCAS_IN.Value
FFRCOPST	RCAS_OUT.Status
FFRCOPVL	RCAS_OUT.Value
FFRESET	RESET
FFROINST	ROUT_IN.Status
FFROINVL	ROUT_IN.Value
FFROOPST	ROUT_OUT.Status
FFROOPVL	ROUT_OUT.Value
FFSEL1ST	SEL_1.Status
FFSEL1VL	SEL_1.Value
FFSEL2ST	SEL_2.Status
FFSEL2VL	SEL_2.Value
FFSEL3ST	SEL_3.Status
FFSEL3VL	SEL_3.Value
FFSPHILM	SP_HI_LIM
FFSPLOLM	SP_LO_LIM

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
FFSPRTDN	SP_RATE_DN
FFSPRTUP	SP_RATE_UP
FFSPSTS	SP.Status
FFSPVAL	SP.Value
FFSTREV	ST_REV
FFTRDIST	TRK_IN_D.Status
FFTRDIVL	TRK_IN_D.Value
FFTRIPST	OUT_TRIP.Status
FFTRIPVL	OUT_TRIP.Value
FFTRVLST	TRK_VAL.Status
FFTRVLVL	TRK_VAL.Value
FFXDDECP	XD_SCALE.DecPt
FFXDS0	XD_SCALE.EU0
FFXDS100	XD_SCALE.EU100
FFXDUIDX	XD_SCALE.UnitIndex
FL	User configured
HIGHAL	None
HIGHALPR	None
IN	Configurable via link object
IND	Configurable via link object
INITREQ	None
INT	User configured
KEYWORD	None
LASTPV	AI: OUT.Value
MODEAPPL	MODE_BLK.Permitted
MODEBLKC	MODE_BLK.Actual + MODE_BLK.Target
MODEPERM	None
MODNAME	VFD model number
MODNUM	None
NAME	None
NN	User configured

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
NODENUM	None
NODETYP	None
NTWKNUM	None
NUMSCHD	MIB: Network Management
OFFNRMFL	ALARM_SUM.Current
OFFNRMPR	DISC_PRI
OP	AO, PID: OUT.Value
OUT	Configurable via link object
OUTD	Configurable via link object
OUTDST	Configurable via link object
OUTST	Configurable via link object
PNTMODTY	None
PNTNODTY	None
PNTSTATE	None
PRILNKMS	NMIB: PrimaryLinkMasterFlagVariable
PRIMMOD	None
PTDESC	None
PTEXECST	None
PTINAL	None
PV	AI: OUT.Value; PD, PID: PV.Value
PVAUTO	AI: OUT.Value
PVAUTOST	AI: OUT.Status
PVCALC	AI: FIELD_VAL.Value
PVEUHI	AI: XD_SCALE.EU0; PID: PV_SCALE.EU100
PVEULO	AI: XD_SCALE.EU0; PID: PV_SCALE.EU0
PVEXEUHI	None
PVEXEULO	None
PVEXHIFL	None
PVEXLOFL	None
PVFL	DI: OUT_D.Value

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
PVFORMAT	AI: OUT_SCALE.DecPt; PID: PV_SCALE.DecPt
PVHHFL	ALARM_SUM.Current
PVHHPR	HI_HI_PRI
PVHIFL	ALARM_SUM.Current
PVHIPR	HI_PRI
PVLLFL	ALARM_SUM.Current
PVLLPR	LO_LO_PRI
PVLOFL	ALARM_SUM.Current
PVLOPR	LO_PRI
PVP	AI: OUT.Value; PID: PV.Value
PVRAW	None
PVSOURCE	None
PVSRCOPT	None
PVSTS	AI: OUT.Status; PID: PV.Status
PVTV	None
PVTVP	None
RCASIN	RCAS_IN
RCASIND	RCAS_IN_D
REDTAG	None
ROUTIN	ROUT_IN
RSSTATE	Resource block: RS_STATE
SCHDVER	MIB: Network Management
SEL1	SEL_1
SEL2	SEL_2
SEL3	SEL_3
SLOTNUM	None
SO	DO: OUT_D.Value
SP	AO, PID: SP.Value
SPFORMAT	AO, PID: PV_SCALE.DecPt
SPP	AO, PID: SP.Value

Table 1 TPS/Fieldbus Parameter Cross Reference

TDC Parameter	FF Parameter Source
STATETXT	None
STR	User configured
TBODIDX	Starting OD index of Transducer Block
TBTYPE	None
UNIT	None
USEVFDNO	None
VENNAME	VFD Vendor name
VFDREF	VFD Reference in VFD_Ref_Entry
VFDTAG	VFD Tag in VFD_Ref_Entry
VWODIDX	Function/Resource Block Header
XBLKDDID	Transducer Block Header
XVWODIDX	Transducer Block Header

1.3 Fieldbus/TPS Cross-Reference

The following table provides a fieldbus to TPS cross-reference listing.

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
AI: FIELD_VAL.Value	PV CALC
AI: OUT.Status	PVAUTOST
AI: OUT.Status	BADPVFL
AI: OUT.Status; PID:	PVSTS
AI: OUT.Value	PVAUTO
AI: OUT.Value	LASTPV
AI: OUT.Value; PD, PID:	PV
AI: OUT.Value; PID:	PVP
AI: OUT_SCALE.DecPt;	PVFORMAT
AI: XD_SCALE.EU0;	PVEULO
AI: XD_SCALE.EU0;	PVEUHI
ALARM_SUM.Current	FFALMCUR
ALARM_SUM.Current	OFFNRMFL
ALARM_SUM.Current	DEVLOFL
ALARM_SUM.Current	DEVHIFL
ALARM_SUM.Current	PVLLFL
ALARM_SUM.Current	PVHIFL
ALARM_SUM.Current	PVHHFL
ALARM_SUM.Current	PVLOFL
AO, PID: OUT.Value	OP
AO, PID:	SPFORMAT
AO, PID: SP.Value	SPP
AO, PID: SP.Value	SP
BKCAL_IN.Status	FFBCINST
BKCAL_IN.Value	FFBCINVL
BKCAL_OUT.Status	FFBCOPST
BKCAL_OUT.Value	FFBCOPVL
Block OD index	BLKODIDX

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
CAS_IN	CASIN
CAS_IN.Status	FFCSINST
CAS_IN.Value	FFCSINVL
CAS_IN_D	CASIND
CAS_IN_D.Status	FFCSIDST
CAS_IN_D.Value	FFCSIDVL
Configurable via link	OUT
Configurable via link	IND
Configurable via link	IN
Configurable via link	OUTDST
Configurable via link	OUTD
Configurable via link	OUTST
DI: OUT_D.Value	PVFL
DISC_PRI	OFFNRMPR
DO: OUT_D.Value	SO
DV_HI_LIM	FFDVHILM
DV_HI_PRI	FFDVHIPR
DV_HI_PRI	DEVHIPR
DV_LO_LIM	FFDVLOLM
DV_LO_PRI	DEVLOPR
DV_LO_PRI	FFDVLOPR
FF_GAIN	FFFFGAIN
FF_SCALE.DecPt	FFFFDECP
FF_SCALE.EU0	FFFFS0
FF_SCALE.EU100	FFFFS100
FF_SCALE.UnitIndex	FFFFUIDX
FF_VAL.Status	FFFFVLST
FF_VAL.Value	FFFFVLVL
Function/Resource Block	VWODIDX
GAIN	FFGAIN
HI_HI_LIM	FFHHLM

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
HI_HI_PRI	FFHHPR
HI_HI_PRI	PVHHPR
HI_LIM	FFHILM
HI_PRI	FFHIPR
HI_PRI	PVHIPR
IN.Status	FFINSTS
IN.Value	FFINVAL
LO_LIM	FFLOLM
LO_LO_LIM	FFLLLM
LO_LO_PRI	PVLLPR
LO_LO_PRI	FFLLPR
LO_PRI	PVLOPR
LO_PRI	FFLOPR
MIB: Network	SCHDVER
MIB: Network	NUMSCHD
MIB: Network	ACTSCHVR
MODE_BLK.Actual	FFMODEA
MODE_BLK.Actual +	MODEBLKC
MODE_BLK.Normal	FFMODEN
MODE_BLK.Permitted	MODEAPPL
MODE_BLK.Permitted	FFMODEP
MODE_BLK.Target	FFMODET
NMIB:	ACTVSCHD
NMIB:	PRILNKMS
None	INITREQ
None	APPL
None	\$OPTOL
None	ALENBST
None	NODETYP
None	MODNUM
None	PNTSTATE

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
None	PTEXECST
None	PTDESC
None	EUDESC
None	UNIT
None	KEYWORD
None	NTWKNUM
None	PNTNODTY
None	CONTCUT
None	PRIMMOD
None	FBTYPE
None	PTINAL
None	REDTAG
None	ALMINFO
None	\$SPTOL
None	PVTV
None	TBTYPE
None	PNTMODTY
None	ASSOCCM
None	PVEXLOFL
None	PVEXEUHI
None	PVEXEULO
None	PVSRCOPT
None	PVEXHIFL
None	NAME
None	PVTV
None	PVSOURCE
None	BADPVPR
None	HIGHALPR
None	\$AUXUNIT
None	MODEPERM
None	DISPTYPE

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
None	BLKDDID
None	SLOTNUM
None	PVRAW
None	ASSOCDSP
None	NODENUM
None	STATETXT
None	ALPRIOR
None	USEVFDNO
None	HIGHAL
None.	DEVSTAT
OUT.Status	FFOUTSTS
OUT.Value	FFOUTVAL
OUT_1.Status	FFOUT1ST
OUT_1.Value	FFOUT1VL
OUT_2.Status	FFOUT2ST
OUT_2.Value	FFOUT2VL
OUT_ACCUM.Status	FFOUTAST
OUT_ACCUM.Value	FFOUTAVL
OUT_D.Status	FFOUTDST
OUT_D.Value	FFOUTDVL
OUT_EXP.Status	FFOEXPST
OUT_EXP.Value	FFOEXPVL
OUT_HI_LIM	FFOPHILM
OUT_LO_LIM	FFOPLOLM
OUT_PTRIP.Status	FFPTRPST
OUT_PTRIP.Value	FFPTRPVL
OUT_REM.Status	FFOREMST
OUT_REM.Value	FFOREMVL
OUT_SCALE.DecPt	FFOPDECP
OUT_SCALE.EU0	FFOPS0
OUT_SCALE.EU100	FFOPS100

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
OUT_SCALE.UnitIndex	FFOPUIDX
OUT_TRIP.Status	FFTRIPST
OUT_TRIP.Value	FFTRIPVL
PID: OUT_SCALE.EU0	CVEULO
PID:	CVEUHI
PV.Status	FFPVSTS
PV.Value	FFPVVAL
PV_SCALE.DecPt	FFPVDECP
PV_SCALE.EU0	FFPVS0
PV_SCALE.EU100	FFPVS100
PV_SCALE.UnitIndex	FFPVUIDX
RATE	FFRATE
RCAS_IN	RCASIN
RCAS_IN.Status	FFRCINST
RCAS_IN.Value	FFRCINVL
RCAS_IN_D	RCASIND
RCAS_IN_D.Status	FFRCIDST
RCAS_IN_D.Value	FFRCIDVL
RCAS_OUT.Status	FFRCOPST
RCAS_OUT.Value	FFRCOPVL
READBACK.Status	FFRBST
READBACK.Value	FFRBVL
READBACK_D.Status	FFRBDST
READBACK_D.Value	FFRBDVL
RESET	FFRESET
Resource block:	RSSTATE
ROUT_IN	ROUTIN
ROUT_IN.Status	FFROINST
ROUT_IN.Value	FFROINVL
ROUT_OUT.Status	FFROOPST
ROUT_OUT.Value	FFROOPVL

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
SEL_1	SEL1
SEL_1.Status	FFSEL1ST
SEL_1.Value	FFSEL1VL
SEL_2	SEL2
SEL_2.Status	FFSEL2ST
SEL_2.Value	FFSEL2VL
SEL_3	SEL3
SEL_3.Status	FFSEL3ST
SEL_3.Value	FFSEL3VL
SP.Status	FFSPSTS
SP.Value	FFSPVAL
SP_HI_LIM	FFSPHILM
SP_LO_LIM	FFSPLOLM
SP_RATE_DN	FFSPRTDN
SP_RATE_UP	FFSPRTUP
ST_REV	FFSTREV
Starting OD index of	TBODIDX
Transducer Block Header	XVWODIDX
Transducer Block Header	XBLKDDID
TRK_IN_D.Status	FFTRDIST
TRK_IN_D.Value	FFTRDIVL
TRK_VAL.Status	FFTRVLST
TRK_VAL.Value	FFTRVLVL
User configured	NN
User configured	FL
User configured	INT
User configured	STR
VFD model number	MODNAME
VFD Reference in	VFDREF
VFD revision number	FFDEVREV
VFD Tag in	VFDTAG

Table 2 Fieldbus/TPS Parameter Cross Reference

FF Parameter Source	TDC Parameter
VFD Vendor name	VENNAME
XD_SCALE.DecPt	FFXDDECP
XD_SCALE.EU0	FFXDS0
XD_SCALE.EU100	FFXDS100
XD_SCALE.UnitIndex	FFXDUIDX

1.4 Supported Function Block (AI, PID, AO)

Parameters per

Analog Input Function Block Parameters

Parameters	ACK_OPTION	LO_LIM
	ALARM_HYS	LO_LO_ALM
	ALARM_SUM	LO_LO_LIM
	ALERT_KEY	LO_LO_PRI
	BLOCK_ALM	LO_PRI
	BLOCK_ERR	LOW_CUT
	CHANNEL	MODE_BLK
	FIELD_VAL	OUT
	GRANT_DENY	OUT_SCALE
	HI_ALM	PV
	HI_HI_ALM	PV_FTIME
	HI_HI_LIM	SIMULATE
	HI_HI_PRI	ST_REV
	HI_LIM	STATUS_OPTS
	HI_PRI	STRATEGY
	IO_OPTS	TAG_DESC
	L_TYPE	UPDATE_EVT
	LO_ALM	XD_SCALE

PID Function Block Parameters

Parameters	ACK_OPTION	LO_LO_ALM
	ALARM_HYS	LO_LO_LIM
	ALARM_SUM	LO_LO_PRI
	ALERT_KEY	LO_PRI
	BAL_TIME	MODE_BLK
	BKCAL_HYS	OUT
	BKCAL_IN	OUT_HI_LIM
	BKCAL_OUT	OUT_LO_LIM
	BLOCK_ALM	OUT_SCALE
	BLOCK_ERR	PV
	BYPASS	PV_FTIME
	CAS_IN	PV_SCALE
	CONTROL_OPTS	RATE
	DV_HI_ALM	RCAS_IN
	DV_HI_LIM	RCAS_OUT
	DV_HI_PRI	RESET
	DV_LO_ALM	ROUT_IN
	DV_LO_LIM	ROUT_OUT
	DV_LO_PRI	SHED_OPT
	FF_GAIN	SP
	FF_SCALE	SP_HI_LIM
	FF_VAL	SP_LO_LIM
	GAIN	SP_RATE_DN
	GRANT_DENY	SP_RATE_UP
	HI_ALM	ST_REV
	HI_HI_ALM	STATUS_OPTS
	HI_HI_LIM	STRATEGY
	HI_HI_PRI	TAG_DESC
	HI_LIM	TRK_IN_D
	HI_PRI	TRK_SCALE
	IN	TRK_VAL
	LO_ALM	UPDATE_EVT
	LO_LIM	

Analog Output Function Block Parameters

Parameters	ALERT_KEY	RCAS_OUT
	BKCAL_OUT	READBACK
	BLOCK_ALM	SHED_OPT
	BLOCK_ERR	SIMULATE
	CAS_IN	SP
	CHANNEL	SP_HI_LIM
	FSTATE_TIME	SP_LO_LIM
	FSTATE_VAL	SP_RATE_DN
	GRANT_DENY	SP_RATE_UP
	IO_OPTS	ST_REV
	MODE_BLK	STATUS_OPTS
	OUT	STRATEGY
	PV	TAG_DESC
	PV_SCALE	UPDATE_EVT
	RCAS_IN	XD_SCALE

1.5 Parameters Per Supported Transducer Block Parameter

Transducer Block Name

- Standard Pressure with Calibration Basic DeviceAccess
- Standard Temperature with Calibration Basic Device Access
- Standard Radar Level with Calibration Basic Device Access
- Standard Flow with Calibration Basic Device Access
- Standard Basic Positioner Valve Basic Device Access
- Standard Advanced Positioner Valve Basic Device Access
- Standard Discrete Positioner Valve Basic Device Access

Following are examples:

Standard Advanced Positioner Valve Basic Device Access

Parameters	ACT_FAIL_ACTION	SERVO_RESET
	ACT_MAN_ID	ST_REV
	ACT_MODEL_NUM	STRATEGY
	ACT_SN	TAG_DESC
	ALER_KEY	TRANSDUCER_DIRECTORY
	BLOCK_ALM	TRANSDUCER_TYPE
	BLOCK_ERR	UPDATE_EVT
	COLLECTION_DIRECTORY	VALUE__SN
	FINAL_POSITION_VALUE	VALUE_MAN_ID
	FINAL_VALUE	VALUE_MODEL_NUM
	FINAL_VALUE_CUTOFF_HI	VALUE_TYPE
	FINAL_VALUE_CUTOFF_LO	XD_CAL_DATE
	FINAL_VALUE_RANGE	XD_CAL_LOC
	MODE_BLK	XD_CAL_WHO
	SERVO_GAIN	XD_ERROR
	SERVO_RATE	

Standard Temperature with Calibration Basic Device Access

Parameters	ALERT_KEY	SENSOR_CAL_DATE
	BLICK_ERR	SENSOR_CAL_LOC
	BLOCK_ALM	SENSOR_CAL_METHOD
	CAL_MIN_SPAN	SENSOR_CAL_WHO
	CAL_POINT_HI	SENSOR_CONNECTION
	CAL_POINT_LO	SENSOR_RANGE
	CAL_UNIT	SENSOR_SN
	COLLECTION_DIRECTORY	SENSOR_TYPE
	MODE_BLK	ST_REV
	MODULE_SN	STRATEGY
	PRIMARY_VALUE	TAG_DESC
	PRIMARY_VALUE_RANGE	TRANSDUCER_DIRECTORY
	PRIMARY_VALUE_TYPE	TRANSDUCER_TYPE
	SECONDARY_VALUE	UPDATE_EVT
	SECONDARY_VALUE_UNIT	XD_ERROR

Standard Pressure with Calibration Basic Device Access

Parameters	ALERT_KEY	SENSOR_CAL_METHOD
	BLOCK_ALM	SENSOR_CAL_WHO
	BLOCK_ERR	SENSOR_FILL_FLUID
	CAL_MIN_SPAN	SENSOR_ISOLATOR_MTL
	CAL_POINT_HI	SENSOR_RANGE
	CAL_POINT_LO	SENSOR_SN
	CAL_UNIT	SENSOR_TYPE
	COLLECTION_DIRECTORY	SECONDARY_VALUE
	MODE_BLK	ST_REV
	PRIMARY_VALUE	STRATEGY
	PRIMARY_VALUE_RANGE	TAG_DESC
	PRIMARY_VALUE_TYPE	TRANSDUCER_DIRECTORY
	SECONDARY_VALUE_UNIT	TRANSDUCER_TYPE
	SENSOR_CAL_DATE	UPDATE_EVT
	SENSOR_CAL_LOC	XD_ERROR

1.6 Schedule Parameters

Link Active Schedule (LAS) Parameters

Table 3 Schedule Summary Parameters

Octet Index	Parameters	Description
1	ScheduleHeader	This is the schedule header and is encoded as 0xFF
2	ScheduleSummary	This attribute identifies the type of schedule transfer SPDU – 0x00 to 0x07. Default is a 0x01.
3-4	ScheduleVersionNumber	The schedule version number. If this field is zero it represents a null schedule.
5-6	ScheduleBuilderIdentifier	The valid range for this variable is 0x0100 to 0x0FFF. This is the source of the schedule and is not used by LAS.
7	NumberOfSubSchedules	This attribute specifies the number of sub-schedules in the schedule.
8	MaximumSchedulingOverhead	This attribute specifies the maximum scheduling overhead, V (MSO), that was used in constructing the schedule. The range is 0x00 to 0x3F. The default value is 0x3F. The unit of this variable is the transmission duration of one octet. This overhead is included in the time allocated for each scheduled activity, and is only used during schedule construction for determination of whether a DLE can serve as LAS for an existing schedule.
9-10	StorageRequiredForTotalSchedule	This attribute specifies the total storage space for the schedule summary and all of the sub-schedules in octets (bytes).
11-12	ScheduleJitterCommitment	This attribute specifies the minimum resolution of the time clock required to execute this schedule and still meet the jitter specifications to which the schedule was constructed. The units are 1/32 millisecond.
13-14	V_MRD_V_ST	This attribute is the value of the maximum response delay slot time multiplied by the octet duration., which was used in the construction of the schedule.
15-16	V_TSL	This attribute is the link-id of the source of time distribution, which was used to specify the DL time used in this schedule.
17-22	T0	This attribute is the DL time base as referenced from the DL time publisher. Not used by LAS
23-26	MacrocycleDuration	This attribute specifies the duration of the link macrocycle. The duration is equal to the least common multiple of periods of all sub-schedules. The clock tick unit is 1/32 of a millisecond.

27-28-	SubScheduleRecordReference(64)	This attribute specifies the sub-schedule identifier. The identifier must not be zero or equal subsequence identifiers. The 1 st zero identifier in this array terminates the Schedule Summary
--------	--------------------------------	---

Table 4 Sub Schedule Parameters

Octet Index	Parameters	Description
1	SubScheduleHeader	This is the sub schedule header and is encoded as 0xFF
2	SubSchedule	This attribute specifies the type of schedule transfer, sub-schedule. The encoding is 0x08 to 0x0F where the value of 0x09 to 0x0F represents the version number of the schedule transfer encoding.
3-4	SubScheduleVersionNumber	This is the version number of the schedule that is identical to the ScheduleVersionNumber in the Schedule Summary.
5-6	SubScheduleIdentifier	This attribute, if non-zero, is the sub-schedule identifier as referenced in the Schedule Summary. If zero, this is the last octet of this SPDU and is used as an end marker for sub-schedules.
7-10	SubSchedulePeriod	This attribute specifies the non-zero execution period of all sequences in the sub-schedule. The clock tick unit is 1/32 of a millisecond.
11-253	<i>Sequences 1-n</i>	<i>Octets 11 to <253 specify one or more sequences as described in the table below. The end of a sub-schedule is indicated by 4 consecutive octets whose value is zero</i>

Table 5 Sequence Parameters

Octet Index	Parameters	Description
1-4	ScheduleStartingTimeOffset	This attribute is used to specify a non-zero starting time offset from the T0 attribute specified in the Schedule Summary. The clock tic units are 1/32 of a millisecond. Since T0 is not used by FF the calculation is as follows: DL-Time –V (DLT0) = V (LST0) + C (NT) = N * (macrocycle-duration) + M * (sub-schedule period) + Schedule-Starting-time offset where, N and M are non-negative integers.
5-6	MaxComputedDurationOfSequence	This attribute is for reference and is not used by LAS. This specifies the worst case duration of the sequence in 1/32 of a millisecond ticks.
7-n	<i>Publisher Elements 1-n</i>	<i>Octets 7 – n specify 1 or more elements as specified in the table below. The end of sequence marker is two octets whose value is zero.</i>

Table 6 Publisher Elements

Octet Index	Publisher Elements	Description
1	ElementType	This attribute specifies the type of transaction. For now the TPSB only provides elements for CD_Request (compel data). The encoding of the bits shall be a CD-request short DLCEP (1011 00PP). The PP is a priority code where: 01 = urgent, 10 = normal, and 11 = time available.
2-3	ElementVariable	This attribute is the short DLCEP-address. The first octet is the node address (14-F7), the second octet is a selector (20-F7) that indicates the published parameter.

1.7 Network Management Parameters

Network Management Directory Entry

The directory entry values provide information about the number of Network Management objects and the address of the objects in the device. This directory is supplied by the manufacturer and is not changeable by a user.

Table 7 Network Management Directory Entry

Directory Entry	Description
DirectoryRevisionNumber	This entry identifies the revision level of the NMA Directory Object.
NumberOfDirectoryObjects	This entry identifies the total number of directory objects in the NMA. The directory objects are consecutively numbered in the Object Dictionary beginning with S-OD index +1.
TotalNumberOfDirectoryEntries	This entry identifies the total number of references, excluding the directory header, in a single directory. Each reference is composed of two unsigned16 array entries. The 1 st reference is the Stack Management reference.
DirectoryIndexOfFirstCompositeListReference	This directory entry is the directory entry number of the 1 st composite list reference. Composite list references contain a directory offset and not an OD Index. The List of MNE composites below is a composite reference.
NumberOfCompositeListReferences	The number of composite list referenced by this directory.
StackMgtOdIndex	The OD Index for the Stack management entity for the NMA.
NumberOfObjectsInStackManagement	The number of objects in the Stack management composite. Usually 1.
VcrListOdIndex	The starting OD Index for the composite for the list of Virtual Communication Relationships accessible through the NMA.

NumberOfObjectsInVcrList	The Number of VCR objects. The 1 st object is the VcrListControlVariable. The 2 nd object is the VCRLISTCharacteristicRecord. The 3 rd object through MaxEntries is the list of VCRStaticEntry objects. The list of VCRDynamicRecords follows the static VCR's. If implemented , a list of VCRStatistic records follow the dynamic VCRs.
DlmeBasicOdIndex	The OD Index for the composite representing the BASIC classes devices that is accessible through the NMA.
NumberOfObjectsInDlIBasic	The number of basic Data Link Management Entity Objects. The 1 st object is the DlmeBasicCharacteristicRecord.
DlmeLinkMasterOdIndex	The OD Index for the composite representing LINK MASTER class device accessible through the NMA. If the device is a BASIC device this entry is set to zero.
NumberOfObjectsInDlILme	The number of objects in the DlmeLinkMaster composite. The 1 st object is the DlmeLinkMaster Capabilities Variable. This entry is zero for a BASIC device.
LinkScheduleListOdIndex	The OD Index for the composite representing the objects for a DlmeLinkSchedule class accessible through the NMA. If the device is a BASIC device this entry is set to zero.
NumberOfObjectsInDlILinkSchedule	The number of objects in the DlmeLinkScheduleList. If the device is a BASIC device this entry is set to zero. The 1 st object is the Link Schedule Activation Variable. The 2 nd object is the Link Schedule Characteristic Record. The 3 rd set of object is the ListOfScheduleDescriptorRecords and the 4 th set of objects is the ListOfScheduleDomains. Assume the starting OD Index for the DlmeLinkSchedule is 100 and there is one schedule domain. The OD Index for that domain is 103. If two schedule domains the OD Index for the 1 st domain would be 104 and the 2 nd would be 105.
DlmeBridgeOdIndex	This OD Index represents the composite for a BRIDGE device. If the data link layer is not a BRIDGE class device, this entry is set to zero.
NumberOfObjectsInDlIBridge	The number of object in the DlmeBridge composite.
PlmeBasicOdIndex	This OD Index entry represents the address for the composite representing the PlmeBasic object(s) (Physical layer basic description).
NumberOfObjectsInPhyLme	Number of objects in the PlmeBasic composite.
ListOfMmeReferences	This entry is a reference to the Media Management composite objects (MmeWireODIndex, NumberO MmeWire objects), that are accessible through the NMA.
NumberOfMmeComposites	This entry represents the number of consecutive MME Composite reference in the list of MME references.
FirstMmeComposite	This represents the OD Index for the MmeWireOdIndex
NumberOfObjectsInMme	This represents the number of MnewireObjects.

Table 8 Network Management Parameters

Network Management Parameters	Parameter Description
DlmeLinkMasterInfo	Data Link Mgmt. Entity Link Master Information. Refer to <i>DlmeLinkMasterInfo</i> Table – below. This attribute is also defined in the IEC/ISA DL Specification Part 4 9.4.1.3 (e.5.i).
DlmeLinkSettings	Data Link Mgmt. Entity Link Settings Refer to Table 9 DlmeLinkSettings.
DlOperateFunctionalClass	DI Operate Functional Class
MaxTokenHoldTimeArray	Refer to Table 12 DlmeLinkMasterInfo.
PlmeBasicInfo	Physical Layer ME Basic Info Refer to Table 13 PlmeBasicInfo.
VcrStaticEntry	Virtual Communication Resource List Refer to Table 11 VcrStaticEntry.

DlmeLinkSettings

The following Table describes the Attributes for **DlmeLinkSettings** (Class) of **Root** (subclass)

Table 9 DlmeLinkSettings

Name	Description
SlotTime	Defined in the IEC/ISA DL Specification Part 4 chapter 5.7.1.1.
PerDlPduPhlOverhead	Defined in the IEC/ISA DL Specification Part 4 chapter 5.7.1.2.
MaxResponseDelay	Defined in the IEC/ISA DL Specification Part 4 chapter 5.7.1.3. Attributes SlotTime and MaxResponseDelay should be set in such a way that the product represents the maximum response delay in octets of the device.
FirstUnpolledNodeId	Defined in the IEC/ISA DL Specification Part 4, clause 5.7.5.15.
ThisLink	ThisLink V(TL) Defined in the IEC/ISA DL Specification Part 4 chapter 5.7.1.9. It is the address of the link that the device is currently connected to.
MinInterPduDelay	MinInterPduDelay V(MD) Defined in the IEC/ISA DL Specification Part 4 chapter 5.7.1.12. It is the minimum inter PDU delay the device requires to operate correctly.

NumConsecUnpolledNodeId	NumConsecUnpolledNodeId V(NUM) Defined in the IEC/ISA DL Specification Part 4, clause 5.7.5.16.
PreambleExtension	Defined in ISA/IEC Part 2 Chapter 6.2. It is an Unsigned8 variable whose allowable values are between 0 and 7.
PostTransGapExtension	Defined in ISA/IEC Part 2 Chapter 6.2. It is an Unsigned8 variable whose allowable values are between 0 and 7.
MaxInterChanSignalSkew	Defined in ISA/IEC Part 2 Chapter 6.2. It is an Unsigned8 variable whose allowable values are between 0 and 7.
Attribute: TimeSyncClass	Defined in IEC/ISA DL Specification Part 4 chapter 5.7.1.25. It represents the TimeSyncClass for the link.

MaxTokenHoldTimeArray

The following Table describes the Attributes for **MaxTokenHoldTimeArray** (Class) of **Root** (subclass)

Table 10 MaxTokenHoldTimeArray

Name	Description
SlotTime	This attribute is an array of 8 OctetString elements. Each element in the array is 64 octets long. Each 64 Octet array element contains 32 values. Each value consists of 2 octets that indicates the initial amount of local link capacity for each active node in the LiveListStatusArrayVariable. The order of values corresponds to the order of nodes in the LiveListStatusArrayVariable. These values are defined in the IEC/ISA DL Specification Part 4 chapter 5.7.5.10. This attribute is represented in FMS as: Defined in the IEC/ISA DL Specification Part 4 chapter 5.7.1.1.

VcrStaticEntry

The following Table describes the Attributes for VcrStaticEntry (Class) . The TPSB automatically builds these entries during building of the LAS Schedule. The table below depicts the binary values that are loaded to the devices.

Table 11 VcrStaticEntry

Name	Description			
FasArTypeAndRole	The semantics of the elements contained in this attribute are defined in the FF FAS specification. It is a OctetString of length 1, defined as follows:			
	B8 B7 B6 B5 (FAS Role) NotConfig: 0000 = 0 Peer Initiator: 1001 = 9 Peer Responder: 0001 = 1 Client: 1010 = 10 Server: 0011 = 3 Source: 0100 = 4 Sink: 0101 = 5 Publisher: 0110 = 6 Subscriber: 0111 = 7	B4 B3 B2 (FAS AR Type) NotConfig: 000 = 0 QUB: 001 = 1 QUU: 010 = 2 BNU: 011 = 3	B1 (lsb) (FMS Bypass) FALSE 0 TRUE: 1 When FMS Bypass is TRUE: 1TRUE, FMS is bypassed (the user application interacts directly with FAS).	
	FasArTypeAndRole Coding The FAS specification defines the valid combinations of AR Type and Role attributes. The definition of the stack, as defined in the StackManagement attribute FasArTypesAndRolesSupported further restricts the values that may be assigned to this attribute. A value of 0x00 indicates that this VCR is unconfigured.			
FasDllLocalAddr	It is defined in the DLL Mapping portion of the FAS specification. It specifies a DLCEP-Address or a DL(SAP)-Address (individual or group). If it is not a group address, then only the selector is defined. The structure of DL-Addresses is defined in the Data Link Layer Protocol Specification, Annex A. NOTE: A value of zero indicates that the relationship is "Fre" in FAS terminology.			
FasDllConfiguredRemoteAddr	This element is defined in the DLL mapping portion of the FAS specification. Depending on the FAS AR Type, it corresponds to either the FAS QUB Remote DLCEP-Address or to the FAS QUU Remote DL(SAP)-Address (individual or group). It is not used for Publisher AREPs. Its value is set to zero to indicate that the FAS RemoteAddressConfigurationType is FREE (for QUU and QUB AREPs). The structure of DL-Addresses is defined in the Data Link Layer Protocol Specification, Annex A. NOTE: A value of zero for QUU and QUB AREPs indicates the relationship endpoint is "Free" in FAS terminology.			
FasDIISDAP	The individual fields of this element are defined in the DLL mapping portion of the FAS specification. This attribute is an OctetString of length 1 defined as follows:			
	B8 LAS Scheduled FALSE 0 TRUE 1	B7 B6 B5 Delivery Features Unordered 000 Ordered 001 Disordered 010 Classical 011	B4 B3 DLPDU Authentication Ordinary 00 Source 10 Maximal 11	B2 B1 (lsb) Priority TimeAvailable 11 Normal 10 Urgent 01
FasDllMaxConfirmDelayOnConnect	It is defined in the DLL mapping portion of the FAS specification as the MaxConfirmDelayOnDIConnect for QUB and BNU AREPs. It is not used for QUU AREPs. It is the maximum amount of time to be used to complete the DL Connection establishment process.			
FasDllMaxConfirmDelayOnData	It defined in the DLL mapping portion of the FAS specification as the MaxConfirmDelayOnDIData for QUB AREPs and as the MaxConfirmDelayOnUnitData for QUU AREPs. It defines the maximum amount of time that the data link layer has to complete individual connectionless and connection-oriented data transfers.			

FasDIIMaxDlsdu Size	It is defined in the DLL mapping portion of the FAS specification as the MaxDlsduSizeFromRequester/MaxDlsduSizeFromResponder for QUB and BNU AREPs and as the MaxDlsduSize for QUU AREPs. It is the maximum permitted Data Link Service Data Unit (DLSDU) size that can be sent on this VCR.			
FasDIIResidual ActivitySupported	It is defined in the DLL mapping portion of the FAS specification. It is used for QUB VCRs only, providing the value for both the ResidualActivityAsSender and ResidualActivityAsReceiver QUB AREP attributes. When TRUE, this Flag indicates that residual activity monitoring will be performed by the DLL in each direction of transfer. The FasDIIMaxConfirmDelayOnConnect indicates the inactivity time-out interval.			
FasDIITimeliness Class	It is composed of publisher and subscriber timeliness classes (defined in the DLL mapping portion of the FAS specification) and the subscriber's Duplicate PDU Detection Supported flag (defined in the BNU AREP attribute in the FAS specification). It is used for BNU VCRs only. It is a OctetString of length 1 octet defined as follows:			
	B8 Reserved	B7 B6 B5 Publisher DL Timeliness Class None: 000 Residence 001 Update 010 Synchronous 011 Transparent 100	B4 Duplicate PDU Detection Supported FALSE 0 TRUE	B3 B2 B1 (Isb) Detection Supported Subscriber DL Timeliness Class (present for subscriber VCR only) None: 000 Residence 001 Update 010 Synchronous 011 Transparent 100
FasDIIPublisher TimeWindowSize	It is defined in the DLL mapping portion of the FAS specification as the PublisherTimeWindowSize. It is used for BNU VCRs only and then only if the PublisherTimelinessClass of this record is not None.			
FasDIIPublisher SynchronizingDlcep	It is defined in the DLL mapping portion of the FAS specification. It is used for BNU VCRs only and then only if the PublisherTimelinessClass (for Publisher AREPs) is Update or Synchronous. This attribute is the DL-address of the DLCEP that is to be used for synchronization within the DLL.			
FasDIISubscriberTimeWindowSize	It is defined in the DLL mapping portion of the FAS specification as the SubscriberTimeWindowSize. It is used for Subscriber BNU VCRs only and then only if the SubscriberTimelinessClass of this record is not None.			
FasDIISubscriberSynchronizingDlcep	It is defined in the DLL mapping portion of the FAS specification as the SubscriberSynchronizingDlcep. It is used for BNU VCRs only and then only if the SubscriberTimelinessClass (for Subscriber AREPs) is Update or Synchronous. This attribute is the DL-address of the DLCEP that is to be used for synchronization within the DLL.			
FmsVfdId	It is defined in the FMS specification as the FMS VFD ID Static VCR Information attribute. It references the VFD associated with this VCR. This value is unambiguous within the scope of the physical device, and must be one of the values contained in the SMIB List of VFD Reference Entry Objects. This attribute is used for QUB Server, QUB Peer, QUU Source and BNU Publisher VCRs. Definition and handling of FMS VFDs are described in the FOUNDATION Fieldbus Specification System Architecture.			
FmsMaxOutstandingServicesCalling	It is defined in the FMS Specification as the MaxOutstandingServicesCalling Static VCR Information attribute. It is used only for QUB VCRs. NOTE: This value may be used by an implementation in the determination of DL max queue depth.			

FmsMaxOutstandingServicesCalled	It is defined in the FMS Specification MaxOutstandingServicesCalledStatic VCR Information attribute. It is used only for QUB VCRs. NOTE: This value may be used by an implementation in the determination of DL max queue depth.
FmsFeaturesSupported	It is defined in the FMS Specification as the FmsFeaturesSupported Static VCR Information attribute. It indicates the services that are supported by FMS for the VCR. It is a BitString of length 8 octets. Each VCR may be defined to support all or a subset of the services specified by this element. The format of this element is defined in Section 3.3.2 of the FMS Specification. This element is not used when FMS is bypassed (see FasArTypeAndRole).

DlmeLinkMasterInfo

The following Table describes the Attributes for **DlmeLinkMasterInfo** (Class)

Table 12 DlmeLinkMasterInfo

Name	Description												
DlmeLinkMasterOdIndex	This attribute specifies the first OD index in the Dlme Link Master composite object, which is the DlmeLinkMasterCapabilitiesVariable.												
DlmeLinkMasterCapabilitiesVariable	This element specifies the capabilities supported by the Link Master. It is represented in FMS as: <table> <tr> <td>FMS Object Name</td><td>DlmeLinkMasterCapabilitiesVariable</td></tr> <tr> <td>Object Type</td><td>Simple Variable</td></tr> <tr> <td>Data Type</td><td>Octet string</td></tr> <tr> <td>Length</td><td>1</td></tr> <tr> <td>FMS Access Right</td><td>RaWa</td></tr> </table> The table below illustrates the structure of this variable. A bit value of 1 indicates that the corresponding attribute is present in the NMIB. A bit value of 0 indicates that the corresponding attribute is not present. <table> <tr> <td> B8 reserved B7 reserved B6 reserved B5 reserved </td><td> B4 reserved B3 LAS Schedule B2 Last Values Record Supported B1 Link Master Statistics Record Supported </td></tr> </table>	FMS Object Name	DlmeLinkMasterCapabilitiesVariable	Object Type	Simple Variable	Data Type	Octet string	Length	1	FMS Access Right	RaWa	B8 reserved B7 reserved B6 reserved B5 reserved	B4 reserved B3 LAS Schedule B2 Last Values Record Supported B1 Link Master Statistics Record Supported
FMS Object Name	DlmeLinkMasterCapabilitiesVariable												
Object Type	Simple Variable												
Data Type	Octet string												
Length	1												
FMS Access Right	RaWa												
B8 reserved B7 reserved B6 reserved B5 reserved	B4 reserved B3 LAS Schedule B2 Last Values Record Supported B1 Link Master Statistics Record Supported												
DlmeLinkMasterInfoRecord	This attribute specifies information of DL Link Master related management. Its attributes are defined in the Dlme Link Master Information Class. This record is writeable for all Link Master capable devices using FMS Write.req. The NMA returns a positive response if no validation is done or if the validation of the data is successful. The new values take effect immediately.												

PrimaryLinkMasterFlagVariable	The default value of this attribute is FALSE. When this flag is set to "TRUE," this node is the primary Link Master. A write to this variable orders the receiving Link Master to try to become the LAS. If the Link Master becomes the LAS, it returns an FMS Write.rsp(+). If this attempt fails, it sends an FMS Write.rsp(-). The behavior is described in the DLL portion of the specification. NOTE: It is a configuration error to have more than one device on a link configured as the Primary Link Master. <table><tr><td>FMS Object Name</td><td>PrimaryLinkMasterFlagVariable</td></tr><tr><td>Object Type</td><td>Simple Variable</td></tr><tr><td>Data Type</td><td>Boolean</td></tr><tr><td>Length</td><td>1</td></tr><tr><td>FMS Access Right</td><td>RaWa</td></tr></table> Element = PrimaryLinkMasterFlagVariable a) The value of this variable may be set to TRUE only through an FMS Write or through a local interface. b) When the value of this variable is TRUE and the device is attached to the network at one of the default addresses, the NMA regards the value as a configuration value and does not attempt to acquire the LAS. c) When the value of this variable is TRUE or when an FMS Write.ind is received to set it to TRUE by the NMA of a Link Master device that is not the LAS and that is attached to the network at an assigned address (not one of the default addresses), the NMA instructs the DLME to acquire the LAS. Two outcomes are possible: - If the DLME indicates that the attempt has failed, the NMA sets the value of this variable to FALSE. If an FMS Write response is pending, the NMA returns a negative response. - If the request succeeds on either the first or the second attempt, the NMA sets the value of this variable to TRUE (if not already TRUE). If an FMS Write response is pending, the NMA returns a positive response.	FMS Object Name	PrimaryLinkMasterFlagVariable	Object Type	Simple Variable	Data Type	Boolean	Length	1	FMS Access Right	RaWa										
FMS Object Name	PrimaryLinkMasterFlagVariable																				
Object Type	Simple Variable																				
Data Type	Boolean																				
Length	1																				
FMS Access Right	RaWa																				
LiveListStatusArrayVariable	This attribute is an OctetString of 32 Octets in which each bit represents a node address on the segment. A value of "1" indicates that the device is communicating on the segment, and a value of "0" indicates that the device is not communicating on the segment. The first bit corresponds to node address 0x00. The last bit corresponds to node address 0xFF. The range of valid node addresses is 0x10 ... 0xFF. It is represented in FMS as: <table><tr><td>FMS Object Name</td><td>PrimaryLinkMasterFlagVariable</td></tr><tr><td>Object Type</td><td>Simple Variable</td></tr><tr><td>Data Type</td><td>OctetString</td></tr><tr><td>Length</td><td>32</td></tr><tr><td>FMS Access Right</td><td>Ra</td></tr></table> The first bit is the first bit that is transmitted and the last bit is the last bit that is transmitted on the bus. For example, for a device with a node address of 16 (decimal) the following would be transmitted on the bus: <table><tr><td>Bit Number</td><td>0 - 7</td><td>8 - 15</td><td>16 – 23</td><td>...</td></tr><tr><td>Octet String</td><td>0x00</td><td>0x00</td><td>0x80</td><td>...</td></tr></table>	FMS Object Name	PrimaryLinkMasterFlagVariable	Object Type	Simple Variable	Data Type	OctetString	Length	32	FMS Access Right	Ra	Bit Number	0 - 7	8 - 15	16 – 23	...	Octet String	0x00	0x00	0x80	...
FMS Object Name	PrimaryLinkMasterFlagVariable																				
Object Type	Simple Variable																				
Data Type	OctetString																				
Length	32																				
FMS Access Right	Ra																				
Bit Number	0 - 7	8 - 15	16 – 23	...																	
Octet String	0x00	0x00	0x80	...																	

MaxTokenHold TimeArray	This attribute is an array of 8 OctetString elements. Each element in the array is 64 octets long. Each 64 Octet array element contains 32 values. Each value consists of 2 octets that indicates the initial amount of local link capacity for each active node in the LiveListStatusArrayVariable. The order of values corresponds to the order of nodes in the LiveListStatusArrayVariable. These values are defined in the IEC/ISA DL Specification Part 4 chapter 5.7.5.10. This attribute is represented in FMS as: FMS Object Name MaxTokenHoldTimeArray Object Type Array Data Type OctetString Length 8 FMS Access Right RaWa
BootOperat FunctionalClass	This attribute specifies how the device should function in the network the next time it boots up. This permits the NMgr to configure a device capable of being a Link Master as a Basic device. Also, a Bridge can be configured to be Link Master or a Basic device. Values are defined in Table 22. A NMgr configures this attribute using FMS write. The NMA returns a positive response to the write if a valid value is written.
CurrentLink SettingsRecord	This record reflects the current settings for the link. Its attributes are defined in the DlmeLinkSettings record. For Link Master devices that are not acting as the LAS, this represents the settings received from the LAS.
ConfiguredLink Settingsrecord	This represents the desired configuration of DLL and PhL management information for the link, This attribute is writeable using FMS Write.req for all Link Master capable devices. The NMA returns a positive response to the write request if no validation is done or if the validation is successful. If the device is or becomes LAS, and if the values are different from the CurrentLinkSettingsRecord, the LAS should clear its live list and reconstruct it using the normal link maintenance procedures. The attributes of this record are defined in the DlmeLinkSettings record.
DlmeLinkMaster StatisticsRecord	This attribute specifies statistics information of DL Link Master related management. Its attributes are defined in the Dlme Link Master Statistics Class in addition to the Dlme Basic Statistics Class. This record is supported if the Link Master Statistics Record Supported bit of the DlmeLinkMasterCapabilitiesVariable is set to TRUE.
LastValuesRecord	This attribute contains information monitored immediately before a NumLasRoleDeleg/Claim/DelegTokenHoldTimeout occurred.

PlmeBasicInfo

The following Table describes the Attributes for **PlmeBasicInfo** (Class)

Table 13 PlmeBasicInfo

Name	Description	
InterfaceMode	This element is defined in ISA/IEC Chapter 6.2, and is coded as described below:	
	B8 reserved B7 reserved B6 reserved B5 reserved	B4 reserved B3 reserved B2 reserved B1 (lsb) Interface Mode Half Duplex 0 Full Duplex 1
LoopBackMode	This element is defined in ISA/IEC Part 2 Chapter 6.2. It is coded as described below:	
	B8 reserved B7 reserved B6 reserved B5 reserved	B4 reserved B3 reserved B2 reserved B1 (lsb) Disabled 00 MAU 01 MDS 10
XmitEnabled	This variable indicates when transmission is enabled on a channel. Transmission is enabled when value is 1.	
	B8 Chan 8 B7 Chan 7 B6 Chan 6 B5 Chan 5	B4 Chan 4 B3 Chan 3 B2 reserved B1 (lsb) Chan 1

RevEnabled	This variable indicates whether or not reception is enabled on a channel. Transmission is enabled when value is 1.	
	B8 Chan 8 B7 Chan 7 B6 Chan 6 B5 Chan 5	B4 Chan 4 B3 Chan 3 B2 reserved B1 (lsb) Chan 1
PreferredReceiveChannel	This element is defined in ISA/IEC Part 2 Chapter 6.2, and is coded as described below:	
	B8 reserved B7 reserved B6 reserved B5 reserved	B4 reserved B3 reserved B2 reserved B1 (lsb) None 000 = 1 Channel 1 0001 = 1 Channel 2 0010 = 2 Channel 3 0011 = 3 Channel 4 0100 = 4 Channel 5 0101 = 5 Channel 6 0110 = 6 Channel 7 0111 = 7 Channel 8 1000 = 8
MediaTypeSelected	This element defines which media type is currently selected. The number corresponds to the position of the specified medium in the MediumAndDataRagesSupported attribute in the DImeBasicCharacteristics class.	
ReceiveSelect	This element defines the channel number that is currently supported. Its values are the same as those defined for the PreferredReceiveChannel attribute.	

System Management Parameters

The directory entry values provide information about the number of System Management objects and the address of the objects in the device. This directory is supplied by the manufacturer and is not changeable by a user.

Table 14 System Management Directory Entry

Directory Entry	Description
DirectoryRevisionNumber	This entry identifies the revision level of the Directory Object (1).
NumberOfDirectoryObjects	This entry identifies the total number of directory objects (1).
TotalNumberOfDirectoryEntries	This entry identifies the total number of references, excluding the directory header, in a single directory (5). Each reference is composed of two unsigned16 array entries. The 1st reference is the SM Agent reference.
DirectoryIndexOfFirstCompositeListReference	This directory entry is the directory entry number of the 1st composite list reference (0). There are none currently defined.
NumberOfCompositeListReferences	Number of composite list references (0). None are defined.
SmAgentStartingOdIndex	The starting OD Index for SM Agent object. The objects are as follows: SMSupport, T1, T2, and T3.
NumberOfSmAgentObjects	The number of SM Agent parameters (4).
SynchAndSchedulingStartingOdIndex	The starting OD Index for the Synch and Scheduling Objects. The objects are as follows: Current_Time, LocalTimeDiff, ApClockSynchInterval, TimeLastRcvd, PrimaryApTimePublisher, TimePublisherAddr, spare, and MacroCycleDuration.
NumberOfSynchAndSchedulingObjects	Number of Synch and Scheduling objects (8).
AddressAssignmentStartingOdIndex	The starting OD Index of the Address Assignment objects: DeviceId (read only), PdTag, Operational_Powerup (read only).
NumberOfAddressAssignmentObject	The number of Address Assignment Objects (3).
VfdListStartingOdIndex	The starting OD Index of the VFD list. This is a list of VFD_REF_ENTRY object that represents the Virtual Field Devices in the Physical device.
NumberOfVfdListObjects	The number of VFD_REF_ENTRY objects in the list. For most basic devices this is (2). One for the Management VFD and one for the Application VFD.
FbScheduleStartingOdIndex	The starting OD Index for the Function Block Schedule Objects. This is a list of FB_START_ENTRY objects.
NumberOfFbScheduleObjects	The number of Function Block Scheduling objects.

System Network Management Parameters

The following table contains the System Network Management Parameters.

Table 15 System Management Parameters

System Management Parameters	Parameter Description
ApClockSyncInterval	AP Clock Synchronization Interval is the interval in seconds between time messages on the link. It is set by system management during node address assignment.
Current_Time	Current Time is the application clock time. It is updated by system management, and the application clock synchronization service is used to keep it coordinated with the application clocks of other devices on the extended link,. Its value represents the number of 1/32 millisecond ticks since January 1, 1972.
FB_Start_Entry	Function Block Schedule It is used to represent an entry in the function block starting schedule. The START_TIME_OFFSET element is in the units of time maintained by the data link layer, each count representing 1/32 of a millisecond. A value of FFFFFFFF(hex) for this element indicates an unused entry. See FF Data Type 91.
LocalTimeDiff	Local Time Difference is used to calculate the local time from CURRENT_TIME. It is specified as the number of (+/-) 1/32 millisecond ticks to add to the clock to obtain the local time.
MacrocycleDuration	Macrocycle Duration is the length of the macrocycle in 1/32 of a millisecond ticks. The maximum value represents a duration of approximately 37 hours. The MacroCycle duration is the least common multiple of the Function Block Schedules durations.
PrimaryApTimePublisher	Primary AP Time Publisher is the node address of the primary time publisher for the local link. It is set by system management during node address assignment. It is a configuration error if the SM_SUPPORT variable of the device specified by this node address does not have the Application Clock Synchronization (Time Publisher) feature bit set.
SMsupport	SM Support indicates which features are supported by the System Management Kernel in this device. The features are mapped to the bits in the bit string.
T1	T1 is the preset value for the SM step timer in 1/32 of a millisecond ticks. The default value is specified by the communications profile for the application area.
T2	T2 is the preset value for the SM set address sequence timer in 1/32 of a millisecond ticks. The default value is specified by the communications profile for the application area.
T3	T3 is the preset value for the set address wait timer in 1/32 of a millisecond ticks. The default value is specified by the communications profile for the application area.

Vfd_Ref_Entry	There will be a number of VFD List Objects entries identifying the VFDs in this device. This list identifies, using values local in scope, the virtual field devices that exist in this physical device. These values are the same as those used for the FmsVfdID attribute of the VcrStaticEntry objects in the NMIB. The VFD tags may be assigned by the user. They may optionally be set with the FMS write service, but the reference values should not be changed and will be verified during the write if this capability is supported. In addition, the server must check that the tag is unique within this list, and that it is not blank (containing all spaces).
---------------	---

1.8 VFD Application Parameters

Table 16 VFD Application Directory

Directory Entry	Description
DirectoryRevisionNumber	Current revision number of the directory structure.
NumberOfDirectoryObjects	The number of directory object. Multiple directory objects are contiguous in the OD. Normally 1.
TotalNumberOfDirectoryEntries	The total number of references, excluding the directory header. The first reference is for Action Objects.
DirectoryIndexOfFirstCompositeListReference	This is the directory index for the composite object after the Domain Entry (17). This is normally the Resource Block Composite Object.
NumberOfCompositeListReferences	The number of Composite List References (3).
OdIndexForStartingActionObject	Starting OD Index for the Action Objects in this VFD.
NumberOfActionObjects	The number of Action Objects.
OdIndexOfTheStartingLinkObjectInTheVfd	Starting OD Index for the Link Objects in this VFD.
NumberOfLinkObjectsInTheVfd	The number of Link Objects allocated to this VFD.
OdIndexOfTheStartingAlertObjectInTheVfd	The starting OD Index of the Alert Objects in this VFD.
NumberOfAlertObjectsInTheVfd	The number Alert Objects in this VFD.
OdIndexOfTheStartingTrendObjectInTheVfd	The starting OD Index for the Trend Object in this VFD.
NumberOfTrendObjectsInTheVfd	The number of Trend Objects in this VFD.
OdIndexOfTheStartingDomainObjectInTheVfd	The starting OD Index for application Domain Objects in this VFD.
NumberOfDomainObjectsInTheVfd	The number of application Domain Objects in this VFD.
DirectoryIndexForTheResourceBlock	Index in this directory for the Resource Block (23).
NumberOfResourceBlocksInTheVfd	The number of resource Blocks in the VFD (always 1).
DirectoryIndexForTheFirstTransducerBlockPointer	Index in this directory for the first Transducer Block OD Index (25).
NumberOfTransducerBlocksInTheVfd	The number of Transducer Blocks in this VFD.
DirectoryIndexForTheFirstFunctionBlockPointer	Index in this directory for the first Function Block OD Index.
NumberOfFunctionBlocksInTheVfd	The number of Function Blocks in this VFD.

2. Introduction

2.1 Purpose

This publication defines the user-visible parameters that exist in the **TPS Builder**. It also provides listings of parameters that are applicable to various HPM point types and algorithms, including fieldbus points. For information on how the parameters are related to each other in terms of point types and algorithms, refer to the *TPS Overview* manual.

2.2 Use of This Publication

Use this publication during configuration and during operation when detailed information about TPS Builder parameters is required.

For use in data point configuration, this publication provides definitions of parameters used in the *TPS Builder*.

For use in process operation, this publication provides information about the Function Blocks that appear for the process data points in the TPS Builder.

2.3 Parameter Definition Format

In this dictionary, the parameter definitions are listed in alphabetical order according to the parameter name, which often exceeds eight characters in length. Parameters in this publication are defined using the general format shown below in the sample **Cal_Min_Span** parameter.

Cal_Min_Span sample parameter

Classification	
Description	The minimum calibration span allowed. Ensures that the two calibrated points (high and low) are not too close
FF Data Type	Float
Range	
Usage	Contained
Length	4
Valid Views	4
Storage	Static
Remarks	

View 4 = Other Static – Information that is normally referenced only during configuration or maintenance and that has a specific value unless changed by an operator or instrument technician.

Some parameters in this dictionary do not have data for Min. Value, Max. Value, Min. Ref, Max. Ref; or View. This is because they do not apply to that particular parameter. When no View is applicable, no view is shown; when more than one view applies to a parameter, all applicable views are shown and described.

Parameter attributes defined

Attribute	Definition
Classification	
Description	Brief description of the parameter
FF Data Type	This entry is the data type that defines how the parameter is viewed by the system. The following data types are used in the Foundation Fieldbus dictionary. Access Permissions Alarm Discrete Alarm Float— Alarm Summary— Bit String— Date— Float— Scaling— Unsigned8 (16 or 32)— Integer8 (16 or 32) Value & Status Float Value & Status Discrete Visible String Octet String Mode
Range	This entry is Min. Value Low bound value for range checking Max. Value High bound value for range checking Min. Ref Reference to parameter which value defines the low bound for range checking Max. Ref Reference to parameter which value defines the high bound for range checking

Usage

Tells how the parameter is used:

Contained— Contained parameters may be used to provide values to block algorithms. Their initial values may be set by the manufacturer or as part of configuration, and they may be adjusted during operation. Also, they may be used to reflect the results of block execution. Linking of input or output parameters to contained parameters is not allowed. The first six parameters of any block are always contained parameters.

Input— Input parameters may be defined for function blocks to allow their values to be obtained through function block links to other input or output parameters. They are structured parameters composed of a value and a status. The data type specified for input parameters is dependent on the data type of its value field. The status field, however, is defined to be identical for all input parameters.

Output— Output parameters may be defined in function blocks for use by other function blocks through function block links. They are structured parameters composed of a value and a status. The data type specified for output parameters is dependent on the data type of its value field. The status field, however, is defined to be identical for all output parameters (and input or contained parameters having the status field).

Length

Number of octets in the value (used for fieldbus)

Valid Views

Fieldbus Views are defined as being 1 through 4. They are defined as follows:

View 1 = Operation Dynamic – Information required by a plant operator to view process control, see alarm conditions, and adjust operating targets.

View 2 = Operations Static – Static information which may be required for display to an operator with dynamic information. This information will only be read once when the associated display is first called up, and refreshed when and if the static update code changes.

View 3 = All Dynamic – Information which is changing in value and may need to be referenced by a detail display which shows everything.

View 4 = Other Static – Information that is normally referenced only during configuration or maintenance and that has a specific value unless changed by an operator or instrument technician.

View 4b = Second View 4

View 4c = Third View 4

Storage	Indicates FieldBus data storage class, which is one of the following: Dynamic— Mixed— Non-Volatile— Static—
Remarks	Additional information about the parameter including helpful hints.

Enumeration Name

This is the Name of the Enumeration type. Please refer to the table of Enumerations at the end of this publication for additional information. The first column of the Enumerations table is sorted alphabetically by the Enumeration Name. First, find the Enumeration name in the Enumeration table, then use your judgment, based on your knowledge of your system, to decide which Value (last column in table) is applicable to your parameter.

2.4 Fieldbus Parameter Types

Fieldbus Parameter Types (Type descriptor in parameters) appear as a numeric followed by its description.

The following table describe fieldbus Parameter Data Types.

Table 17 Fieldbus Parameter Data Types

FF Parameter Type	Description
1	Logical (Boolean)
2	Integer8
3	Integer16
4	Integer32
5	Unsigned8
6	Unsigned16
7	Unsigned32
8	Float
9	Visible String
10	Octet String
11	Date
12	Time Of Day
13	Time Difference
14	Bit String
21	Time Value
64	Block Header
65	Value & Status - Floating Point
66	Value & Status - Discrete
67	Value & Status - Bitstring
68	Scaling
69	Mode
70	Access Permissions
71	Alarm Float
72	Alarm Discrete
73	Event Update
74	Alarm Summary

Table 17 Fieldbus Parameter Data Types

FF Parameter Type	Description
75	Alert Analog
76	Alert Discrete
77	Alert Update
78	Trend Float
79	Trend Discrete
80	Trend Bit String
81	FB Link
82	Simulate - Floating Point
83	Simulate - Discrete
84	Simulate - Bitstring
85	Test
86	Action
90	VFD Reference Entry
91	FB Start Entry
97	VCR Static Entry
104	DLME Link Master Information
109	PLME Basic Information
111	DLME Link Settings

2.5 Fieldbus Structured Data Types

This section presents the Simple Variable data types defined by Foundation Fieldbus.

Value & Status - Floating Point Structure

This data structure consists of the value and status of floating point parameters that are Inputs or Outputs.

Data Type: Value & Status - Floating Point

Key Attribute: Index = 65

Attribute: Number of Elements = 2

Attribute: List of Elements

Element	Element Name	Data Type (Index)	Size
1	Status	Unsigned8 - (5)	1
2	Value	Float - (8)	4

Value & Status - Discrete Structure

This data structure consists of the value and status of discrete value parameters.

Data Type: Value & Status - Discrete

Key Attribute: Index = 66

Attribute: Number of Elements = 2

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	Status	Unsigned8 - (5)	1
2	Value	Float - (8) – (5)	1

Scaling Structure

This data structure consists of the static data used to scale floating point values for display purposes.

Data Type: Scaling

Key Attribute: Index = 68

Attribute: Number of Elements = 4

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	EU at 100%	Float - (8)	4
2	EU at 0%	Float - (8)	4
3	Units Index	Unsigned16 - (6)	2
4	Decimal Point	Integer8 - (2)	1

Mode Structure

This data structure consists of bit strings for target, actual, permitted, and normal modes.

Data Type: Mode

Key Attribute: Index = 69

Attribute: Number of Elements = 4

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	Target	Bitstring- (14)	1
2	Actual	Bitstring - (14)	1
3	Permitted	Bitstring - (14)	1
4	Normal	Bitstring - (14)	1

Access Permissions

This data structure consists of access control flags for access to block parameters.

Data Type: Access Permissions

Key Attribute: Index = 70

Attribute: Number of Elements = 2

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	Grant	Bit String - (14)	1
2	Deny	Bit String - (14)	1

Alarm Float Structure

This data structure consists of data that describes floating point alarms.

Data Type: Alarm Float

Key Attribute: Index = 71

Attribute: Number of Elements = 5

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	Unacknowledged	Unsigned8 - (5)	1
2	Alarm State	Unsigned8 - (5)	1
3	Time Stamp	Time Value - (21)	8
4	Normal	Unsigned16 - (6)	2
5	Subcode	Unsigned16 - (6)	4

Event Update Structure

This data structure consists of data that describes a static revision alarm.

Data Type: Event Update

Key Attribute: Index = 73

Attribute: Number of Elements = 5

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	Unacknowledged	Unsigned8 - (5)	1
2	Update State	Unsigned8 - (5)	1
3	Time Stamp	Time Value - (21)	8
4	Static Revision	Unsigned16 - (6)	2
5	Relative Index	Unsigned16 - (6)	2

Alarm Summary Structure

This data structure consists of data that summarizes 16 alerts.

Data Type: Alarm Summary

Key Attribute: Index = 74

Attribute: Number of Elements = 4

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	Current	Bit String - (14)	2
2	Unacknowledged	Bit String - (14)	2
3	Unreported	Bit String - (14)	2
4	Disabled	Bit String - (14)	2

FB Link Structure

This data structure consists of function block link data.

Data Type: FB Link

Key Attribute: Index = 81

Attribute: Number of Elements = 5

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
1	Local Index	Unsigned16 - (6)	2
2	VCR Number	Unsigned16 - (6)	2
3	Sample Type	Unsigned8 - (5)	2
4	Service Operation	Unsigned8 - (5)	1
5	Stale Count Limit	Unsigned8 - (5)	1

Test Structure

This data structure consists of function block test read/write data.

Data Type: Test

Key Attribute: Index = 85

Attribute: Number of Elements = 15

Attribute: List of Elements (shown below)

Element	Element Name	Data Type (Index)	Size
---------	--------------	-------------------	------

1	Value 1	Boolean - (1)	1
2	Value 2	Integer8 - (2)	1
3	Value 3	Integer16 - (3)	2
4	Value 4	Integer32 - (4)	4
5	Value 5	Unsigned8 - (5)	1
6	Value 6	Unsigned16 - (6)	2
7	Value 7	Unsigned32 - (7)	4
8	Value 8	Floating Point - (8)	4
9	Value 9	VisibleString - (9)	32
10	Value 10	Octet String - (10)	32
11	Value 11	Date - (11)	7
12	Value 12	Time of Day - (12)	6
13	Value 13	Time Difference - (13)	6
14	Value 14	Bitstring - (14)	2
15	Value 15	Time Value- (21)	8

VFD Preference Entry Structure

The VFD_REF_ENTRY data type structure is used to represent an entry in the virtual field device reference list.

Numeric Identifier 90

Number of Elements 2

List of Elements

 Data Type Index: 7 (Unsigned32)

 Length: 4

 Data Type Index: 9 (Visible String)

 Length: 32

FB Start Entry Structure

The FB_START_ENTRY data type structure is used to represent an entry in the function block starting schedule. The START_TIME_OFFSET element is in the units of time maintained by the data link layer, each count representing 1/32 of a millisecond. A value of FFFFFFFF(hex) for this element indicates an unused entry.

Numeric Identifier	91
Number of Elements	3
<u>List of Elements</u>	
Data Type Index:	7 (Unsigned32)
Length:	4
Data Type Index:	6 (Unsigned16)
Length:	2
Data Type Index:	7 (Unsigned32)
Length:	4

3. Fieldbus Parameter Definitions (A through Y)

This section contains descriptions of the Fieldbus Foundation parameters. The parameter descriptions are listed in alphabetical order according to the parameter name.

4. A

4.1 Accept_Alm

Classification	Fieldbus Device
Description	Generated when the Accept_D switch is used.
FF Data Type	Alarm Discrete
Range	N/A
Usage	Contained
Length	13
Valid View(s)	N/A
Storage	Dynamic

4.2 Accept_D

Classification	Fieldbus Device
Description	Causes PV_D to become a copy of Out_D.
FF Data Type	Unsigned8
Range	0 Off 1 Accept
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Dynamic

4.3 Accept_Pri

Classification	Fieldbus Device
Description	Priority of accept alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1

Valid View(s)	4
Storage	Static

4.4 Ack_Option

Classification	Fieldbus Device
Description	Defines of whether alarms associated with the function block will be automatically acknowledged.
FF Data Type	Bit String
Range	0 Disable/Enable Discrete Alarm 1 Disable/Enable High High Alarm 2 Disable/Enable High Alarm 3 Disable/Enable Low Low Alarm 4 Disable/Enable Low Alarm 5 Disable/Enable Deviation High Alarm 6 Disable/Enable Deviation Low Alarm 7 Disable/Enable Block Alarm 8 Disable/Enable Alarm 8 9 Disable/Enable Alarm 9 10 Disable/Enable Alarm 10 11 Disable/Enable Alarm 11 12 Disable/Enable Alarm 12 13 Disable/Enable Alarm 13 14 Disable/Enable Alarm 14 15 Disable/Enable Alarm 15
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

4.5 Act_Fail_Action

Classification	Fieldbus Device
Description	Specifies the action the actuator takes in case of failure.
FF Data Type	Unsigned8
Range	0 Undefined 1 SelfClosing 2 SelfOpening 3 HoldLastValue 4 MaximumValue 5 MinimumValue 255 Indeterminate
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

4.6 Act_Man_Id

Classification	Fieldbus Device
Description	The actuator manufacturer's identification number as defined by FOUNDATION Fieldbus.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Non-Volatile

4.7 Act_Mod_Num

Classification	Fieldbus Device
Description	Defines the actuator's model number.
FF Data Type	Visible String
Range	N/A
Usage	Contained

Length	32
Storage	Non-Volatile
Valid View(s)	4

4.8 Act_Sn

Classification	Fieldbus Device
Description	Defines the actuator's serial number.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Storage	Non-Volatile
Valid View(s)	4

4.9 Alarm_Hys

Classification	Fieldbus Device
Description	Determines the amount that the PV must return within the alarm limits before the alarm condition clears. Alarm hysteresis expressed as a percent of the span of the PV.
FF Data Type	Float
Range	0-50 %
Usage	Contained
Length	4
Storage	Static
Valid View(s)	4

4.10 Alm_Rate_Dn

Classification	Fieldbus Device
Description	Defines the ramp rate of change of upward alarm expansion.
FF Data Type	Float
Range	Any Positive Value
Usage	Contained
Length	4
Valid View(s)	4

Storage	Static
---------	--------

4.11 Alm_Rate_Up

Classification	Fieldbus Device
Description	Defines the ramp rate of change of downward alarm expansion.
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

4.12 Alarm_Sum

Classification	Fieldbus Device
Description	Specifies the current alert status, unacknowledged states, unreported states, and disabled states of the alarms associated with the function block.
FF Data Type	Alarm Summary
Range	N/A
Usage	Contained
Length	8
Valid View(s)	1, 4
Storage	Mixed

4.13 Alert_Key

Classification	Fieldbus Device
Description	Defines the identification number of the plant unit. This information may be used in the host for sorting alarms, etc.
FF Data Type	Unsigned8
Range	1-255
Usage	Contained
Length	1
Valid View(s)	4

Storage

Static

4.14 Arith_Type

Classification	Fieldbus Device
Description	Specifies the ID number of the arithmetic algorithm.
FF Data Type	Unsigned8
Range	1 FlowCompensationLinear
	2 FlowCompensationSQRoot
	3 FlowCompensationAppx
	4 BTUFlow
	5 MultiplyDivide
	6 Average
	7 Summer
	8 ForthOrderPoly
	9 HTGCompensatedLevel
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

4.15 Auto_Cycle

Classification	Fieldbus Device
Description	When true, resets time to start of first step.
FF Data Type	Unsigned8
Range	N/A
Usage	Contained
Length	1
Valid View(s)	2
Storage	Static

5. B

5.1 Bal_Time

Classification	Fieldbus Device
Description	Specifies the time (in seconds) for the internal working value of bias or ratio to return to the operator set bias or ratio. In the PID block, it may be used to specify the time constant at which the integral term will move to obtain balance when the output is limited and the mode is Auto, Cas, or RCas.
FF Data Type	Float
Range	positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

5.2 Bias

Classification	Fieldbus Device
Description	Specifies the bias value used in computing the function block output, expressed in engineering units.
FF Data Type	Float
Range	
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Non-Volatile
Reference	OUT_SCALE +/- 10%

5.3 Bias_In_1 - Bias_In_3

Classification	Fieldbus Device
Description	Specifies the value of the constant to be added to IN_1, IN_2, and IN_3, respectively.
FF Data Type	Float
Range	N/A
Usage	Contained

Length	5
Valid View(s)	4
Storage	Static

5.4 Binary_Cl

Classification	Fieldbus Device
Description	Specifies the priority discrete setpoint that commands the actuator closed.
FF Data Type	Value & Status Discrete
Range	0-50
Usage	Input
Length	2
Valid View(s)	4
Storage	Non-Volatile
Reference	OUT_SCALE +/- 10%

5.5 Binary_Op

Classification	Fieldbus Device
Description	Specifies the priority discrete setpoint that commands the actuator open.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	4
Storage	Non-Volatile

5.6 BkCal_Hys

Classification	Fieldbus Device
Description	Specifies the amount that the block output must change away from its output limit before the limit status is turned off, expressed as a percent of span of the output.
FF Data Type	Float
Range	0-50%
Usage	Contained

Length	4
Valid View(s)	4
Storage	Static

5.7 BkCal_In

Classification	Fieldbus Device
Description	The value and status used for backwards tracking of the output, provided by a link to a downstream block's back calculation output parameter.
FF Data Type	Value & Status - Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	3
Storage	Non-Volatile

5.8 BkCal_In_1 - BkCal_In_2

Classification	Fieldbus Device
Description	The back-calculated input value and status required for a splitter.
FF Data Type	Value & Status - Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	4
Storage	Non-Volatile

5.9 BkCal_In_D

Classification	Fieldbus Device
Description	The back-calculated input required for a control block.
FF Data Type	Value & Status - Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	4

Storage	Non-Volatile
----------------	--------------

5.10 BkCal_Out

Classification	Fieldbus Device
Description	The output value and status provided to an upstream block for output tracking when the loop is broken or limited, as determined by the status bits. This information is used to provide bumpless transfer to closed loop control and to prevent windup under limited conditions when that becomes possible.
FF Data Type	Value & Status - Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	3
Storage	Dynamic

5.11 BkCal_Out_D

Classification	Fieldbus Device
Description	The output value and status provided to an upstream block output tracking when the loop is broken, as determined by the status bits. This information is used to provide bumpless transfer to closed loop control.
FF Data Type	Value & Status - Discrete
Range	N/A
Usage	Output
Length	2
Valid View(s)	3
Storage	Dynamic

5.12 BkCal_Sel_1 - BkCal_Sel_3

Classification	Fieldbus Device
Description	Control selector output value and status associated with SEL_1, SEL_2, or SEL_3 input, which is provided as the back calculation input to the upstream block that is linked to SEL_1, SEL_2, or SEL_3, respectively.
FF Data Type	Value & Status - Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	3
Storage	Dynamic

5.13 Block_Alm

Classification	Fieldbus Device
Description	Defines the block alarm that is used for all configuration, hardware, connection failure or system problems in the block. The cause of the alert is entered in the subcode field. The first alert to become active will set the active status in the Status attribute. As soon as the unreported status is cleared by the alert reporting task, another block alert may be reported without clearing the active status, if the subcode has changed.
FF Data Type	Alarm Discrete
Range	N/A
Usage	Contained
Length	13
Valid View(s)	N/A
Storage	Dynamic

5.14 Block_Execution_Sequence

Classification	TPSB Only
Description	Specifies the blocks execution sequence within a control loop. Used in function block scheduling.
FF Data Type	Integer
Range	0-119
Usage	Contained
Length	4

Valid View(s)	N/A
Storage	N/A

5.15 BlockName

Classification	TPSB Only
Description	Defines the block name for Meta Data.
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

5.16 BlockNo

Classification	TPSB Only
Description	Defines the transducer block number in a VFD or function block number in a VFD.
FF Data Type	N/A
Range	0-100
Usage	N/A
Length	N/A
Valid View(s)	N/A
Storage	N/A

5.17 Block_Err

Classification	Fieldbus Device
Description	Reflects the error status associated with the hardware or software components that are associated with a block. It is a bit string, so that multiple errors may be shown.
FF Data Type	Bit String
Range	0 Other 1 BlockConfiguration 2 LinkConfiguration 3 SimulationActive 4 LocalOverride

	5 DeviceFaultState
	6 DeviceMaintenance
	7 InputFailure
	8 OutputFailure
	9 MemoryFailure
	10 LostStaticData
	12 ReadbackCheck
	13 MaintenanceNeeded
	14 PowerUp
	15 OutOfService
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

5.18 Block_Index

Classification	CFF File
Description	Identifies the starting object dictionary index of a block.
FF Data Type	Unsigned16
Range	0-16384
Usage	N/A
Length	2
Valid View(s)	N/A
Storage	N/A

5.19 Block_Tag

Classification	Fieldbus Device
Description	Specifies the tag name of the block. The tag name is to be unique throughout the control system on one link.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	N/A

Storage	N/A
---------	-----

5.20 Body_Temp

Classification	Fieldbus Device
Description	The instrument's body temperature value and status available to the function block.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	3
Storage	Dynamic

5.21 Body_Temp_Unit

Classification	Fieldbus Device
Description	The instrument's engineering units code for the body temperature.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

5.22 Block_Type

Classification	CFF File
Description	Specifies one of the FF block types. For example, AI, PID, AO, and etc. The manufacturer names custom block types
FF Data Type	Visible String
Range	N/A
.	
Usage	Contained
Length	32
Valid View(s)	N/A
Storage	N/A

5.23 ByPass

Classification	Fieldbus Device
Description	The normal control algorithm may be bypassed through this parameter. When bypass is set, the setpoint value (in percent) will be directly transferred to the output. To prevent a bump on transfer to/from bypass, the setpoint will automatically be initialized to the output value or process variable, respectively, and the path-broken flag will be set for one execution.
FF Data Type	Unsigned8
Range	1 Off 2 On
Usage	Contained
Length	1
Valid View(s)	2
Storage	Static

6. C

6.1 Cal_Min_Span

Classification	Fieldbus Device
Description	The minimum calibration span allowed. Ensures that the two calibrated points (high and low) are not too close.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.2 Cal_Point_Hi

Classification	Fieldbus Device
Description	The highest calibrated value on a transducer block.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

6.3 Cal_Point_Lo

Classification	Fieldbus Device
Description	The lowest calibrated value on a transducer block.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

6.4 Cal_Propagate_Vel

Classification	Fieldbus Device
Description	The calibrated propagation velocity of the radar pulse.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Non-Volatile

6.5 Calc_Errrs

Classification	Fieldbus Device
Description	Displays the types of calibration errors.
FF Data Type	Bit String
Range	0 Non-SpecificError 1 DivideByZero 2 Overflow 3 FunctionError 4 FunctionCallNotSupported 5 FuntionCallArgumentCount 6 FunctionCallArgumentType 7 IllegalOperator 8 JMP-OutOfList 9 InvalidDataType 10 TypeConversionFailed 11 ST-Disallowed 12 StackOverflow 13 INST_EX_LIM-Count
Usage	Contained
Length	2
Valid View(s)	4
Storage	Dynamic

6.6 Cal_Unit

Classification	Fieldbus Device
-----------------------	-----------------

Description	The device-description engineering-units-code index for the calibration values.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

6.7 Cas_In

Classification	Fieldbus Device
Description	The remote setpoint value, which must come from another fieldbus block, or from an HPM point through an output connection.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

6.8 Cas_In_D

Classification	Fieldbus Device
Description	The remote setpoint value, which must come from another discrete fieldbus block, or from an HPM point through an output connection.
FF Data Type	Value & Status - Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

6.9 Cfm_Act1_Time - Cfm_Act2_Time

Classification	Fieldbus Device
Description	The maximum time allowed to confirm the first active state

and second active state, respectively.

FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.10 Cfm_Pass_Time

Classification	Fieldbus Device
Description	The maximum time allowed to confirm the passive state.
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.11 Channel

Classification	Fieldbus Device
Description	The number of the logical hardware channel that is connected to this I/O block. This information defines the transducer to be used to interface with the physical world. The I/O block may not transition from OOS mode unless a non zero channel number is entered.
FF Data Type	Unsigned16
Range	1-to Mfg. Limit
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

6.12 Channel_Map

Classification	Fieldbus Device
Description	An optional parameter that aids in the correlation of channels in a device to channels as defined for a plant or

	process.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A
Storage	Static

6.13 CheckBack

Classification	Fieldbus Device
Description	Displays status of final control element.
FF Data Type	Bit String
Range	0 Stat_Safeguard_OP 1 Stat_Safeguard_CL 2 Stat_Binary_OP 3 Stat_Binary_CL 4 Stat_Disc 5 Stat_Disc_D_Op 6 Stat_Disc_D_CL 7 Stat_Rb_D_OP 8 Stat_Rb_D_CL 9 Stat_Torque_D_OP 10 Stat_Torque_D_CL 11 Stat_Sub_Mode 12 Stat_Trav_Time 13 Reserved1 14 Reserved2
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

6.14 Clock_Per

Classification	Fieldbus Device
-----------------------	-----------------

Description	Elapsed time (in hours) of periodic reset.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.15 Clr_Fstate

Classification	Fieldbus Device
Description	Writing a Clear to this parameter will clear the device faultstate state provided the field condition, if any, has cleared.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Off 2 Clear
Usage	Contained
Length	1
Valid View(s)	N/A
Storage	Dynamic

6.16 Collection_Directory

Classification:	Fieldbus Device
Description	A directory that specifies the number, starting indices, and device-description item IDs of the data collections in each transducer within a transducer block.
FF Data Type	Unsigned32
Range:	N/A
Usage:	Contained
Length:	4
Valid View(s):	N/A
Storage:	Static

6.17 Comb_Type

Classification	Fieldbus Device
Description	Determines how multiple IN_D values are combined in the timer function.
FF Data Type	Unsigned8
Range	0 AND 1 OR 2 ANY2 3 ANY3 21 EXACTLY1 22 EXACTLY2 23 EXACTLY3 40 EVEN 41 ODD
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

6.18 Comp_Hi_Lim

Classification	Fieldbus Device
Description	Defines the high limit for the PV compensation term.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.19 Comp_Lo_Lim

Classification	Fieldbus Device
Description	Defines the low limit for the PV compensation term.
FF Data Type	Float
Range	N/A

Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.20 Cond_Time

Classification	Fieldbus Device
Description	Length of conditioning time (in seconds).
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.21 Cond_Type

Classification	Fieldbus Device
Description	Defines the type of conditioning applied to PV_D in the discrete alarm function.
FF Data Type	Unsigned8
Range	0 None 1 Delay 2 Extend 3 Debounce 4 Pulse 5 RT_Pulse
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

6.22 Confirm_Time

Classification	Fieldbus Device
Description	The minimum time between retries of alert reports.

FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

6.23 Control_Opts

Classification	Fieldbus Device
Description	Options which the user may select to alter the calculations done in one of the following control block options: CS, PD, RA, BG, PID, or ML.
FF Data Type	Bit String
Range	0 Bypass_Enable 1 SP-PV_Track_Man 2 SP-PV_Track_Rout 3 SP-PV_Track_LO-IMan 4 SP_Track_retain 5 Direct_Acting 6 Reserved 7 Track_Enable 8 Track_in_Manual 9 PV_for_BkCal_Out 10 Bias_may_be_adjusted 11 Use_BKCAL_OUT_with_IN 12 Restrict_SP_to_limits_in_Cas_and_RCas 13 No_output_limits_in_Man
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

6.24 ControlType

Classification	TPSB Only
-----------------------	-----------

Description	Defines the type of control. This parameter is used by the TPSB internally to distinguish different types of strategies. For example, a function block, a transducer block, a device, and a VFD application are different control types.
FF Data Type	N/A
Range	N/A
Usage	N/A
Length	N/A
Valid View(s)	N/A
Storage	N/A

6.25 Crack_Time

Classification	Fieldbus Device
Description	The maximum time (in seconds) allowed to drop previous confirm.
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

6.26 Crack_Timer

Classification	Fieldbus Device
Description	Shows time (in seconds) taken to drop the confirm.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Dynamic

6.27 Curve_X

Classification	Fieldbus Device
Description	Defines X coordinate of curve.

FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	Static

6.28 Curve_Y

Classification	Fieldbus Device
Description	Defines Y coordinate of curve.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	Static

6.29 Cycle_Sel

Classification	Fieldbus Device
Description	Used to select the block execution method for this block.
FF Data Type	Bit String
Range	0 Scheduled 1 Block_Execution 2 Manuf_Specific
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

6.30 Cycle_Type

Classification:	Fieldbus Device
Description	Identifies the block execution methods available for this resource.
FF Data Type	Bit String

C**Cycle_Type**

Range: 0 Scheduled
 1 Block_Execution
 2 Manuf_Specific

Usage: Contained

Length: 2

Valid View(s): 4

Storage: Static

7. D

7.1 DC_State

Classification	Fieldbus Device
Description	Displays the current state of the device control function.
FF Data Type	Unsigned8
Range	0 ConfirmedPassive 1 ConfirmedState_1 2 ConfirmedState_2 3 GoingToPassive 4 GoingToState_1 5 GoingToState_2 6 FailedPassive 7 FailedState_1 8 FailedState_2 9 Tripped 10 LostInterlock 11 Locked 12 LostMCCPower
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Dynamic

7.2 DD_ITEM

Classification	Fieldbus Device
Description	A unique number which identifies the description of the object that is developed as part of its device description.
FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	N/A

Storage	N/A
---------	-----

7.3 DD_MEMBER

Classification	Fieldbus Device
Description	A unique number which identified the function of the block that is developed as part of its device description.
FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	N/A

7.4 DD_RESOURCE

Classification	Fieldbus Device
Description	Identifies the tag of the resource that contains the device description.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	N/A
Storage	Static

7.5 DD_REV

Classification	Fieldbus Device
Description	Revision of the device description associated with the resource - used by an interface device to locate the device description file for the resource.
FF Data Type	Unsigned16
Range	N/A
Usage	
Length	2
Valid View(s)	N/A
Storage	Static

7.6 DD_REVIS

Classification	Fieldbus Device
Description	The device description revision number assigned to this block.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A
Storage	N/A

7.7 Dead_Band

Classification	Fieldbus Device
Description	Interval where changes in input do not affect output
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

7.8 Dead_Time

Classification	Fieldbus Device
Description	Delay time (in seconds) for the input value.
FF Data Type	Value & Status Floating Point
Range	0-213900000
Usage	Input
Length	5
Valid View(s)	4
Storage	Non-Volatile

7.9 Delay_Time

Classification	Fieldbus Device
-----------------------	-----------------

Description	Specifies the time (in seconds) to delay a change in SP_D to active.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

7.10 Delay_Timer

Classification	Fieldbus Device
Description	Shows the time (in seconds) of several delay functions.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Dynamic

7.11 Description

Classification	CFF File
Description	Fieldbus device description used in the CFF Value File.
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

7.12 Dev_Id

Classification	Fieldbus Device
Description	Device identifier that is universally unique.
FF Data Type	Visible String
Range	N/A

Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

7.13 Dev_Rev

Classification	Fieldbus Device
Description	Manufacturer revision number associated with the resource - used by an interface device to locate the device-description file for the resource.
FF Data Type	Unsigned16
Range	N/A
Usage	N/A
Length	2
Valid View(s)	N/A
Storage	N/A

7.14 Dev_Type

Classification	Fieldbus Device
Description	Manufacturer's model number associated with the resource - used by interface devices to locate the device-description file for the resource.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

7.15 Device_Opts

Classification	Fieldbus Device
Description	Options used in processing the device control function.
FF Data Type	Bit String
Range	0 Interlock 1 TrackSPInLO

	2 Permissive
	3 ResetRequired
	4 Trip
	5 LostPower
	6 PassiveWhenConfigured
	7 BusRestartPassive
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

7.16 DeviceClass

Classification	CFF File
Description	Defines the device functional class. This parameter is resides only in the Capabilities or Value file.
FF Data Type	Unsigned8
Range	0 Undefined 1 Basic 2 Linkmaster 3 Bridge
Usage	N/A
Length	1
Valid View(s)	N/A
Storage	N/A

7.17 Disable_1 - Disable_4

Classification	Fieldbus Device
Description	The disable or use input switch for the respective input selector function.
FF Data Type	Value & Status Discrete
Range	0 Use 1 Disable
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

7.18 Disc_Alm

Classification	Fieldbus Device
Description	The status and time stamp associated with the discrete alarm.
FF Data Type	Alarm Discrete
Range	N/A
Usage	Contained
Length	13
Valid View(s)	N/A
Storage	Dynamic

7.19 Disc_Cond

Classification	Fieldbus Device
Description	Defines the condition of Out_D, the primary discrete output value.
FF Data Type	Unsigned8
Range	N/A
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

7.20 Disc_Lim

Classification	Fieldbus Device
Description	Defines the state of the discrete input which will generate an alarm.
FF Data Type	Unsigned8
Range	PV State
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

7.21 Disc_Pri

Classification	Fieldbus Device
Description	Defines the priority of the discrete alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

7.22 DV_Hi_Alm

Classification	Fieldbus Device
Description	Displays the current status and time stamp associated with the high deviation alarm.
FF Data Type	Alarm Float
Range	N/A
Usage	Contained
Length	16
Valid View(s)	N/A
Storage	Dynamic

7.23 DV_Hi_Lim

Classification	Fieldbus Device
Description	Defines the value of the high deviation alarm limit in engineering units.
FF Data Type	Float
Range	0- to PV Span, +INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

7.24 DV_Hi_Pri

Classification	Fieldbus Device
Description	Defines the priority of the high deviation alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

7.25 DV_Lo_Alm

Classification	Fieldbus Device
Description	Displays the status and time stamp associated with the low deviation alarm.
FF Data Type	Alarm Float
Range	N/A
Usage	Contained
Length	16
Valid View(s)	N/A
Storage	Dynamic

7.26 DV_Lo_Lim

Classification	Fieldbus Device
Description	Defines the value of the low deviation alarm limit in engineering units.
FF Data Type	Float
Range	- PV Span to 0, -INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

7.27 DV_Lo_Pri

Classification	Fieldbus Device
Description	Defines the priority of the low deviation alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

8. E

8.1 Execution_Period

Classification	Fieldbus Device
Description	The time (in 1/32 milliseconds) between initiation of block execution - for periodic execution
FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	N/A

8.2 ElementName

Classification	TPSB Only
Description	Defines the name of the element strategy associated with the LAS Schedule.
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

8.3 En_Close

Classification	Fieldbus Device
Description	When active, allows close commands
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	4
Storage	Non-Volatile

8.4 En_Open

Classification	Fieldbus Device
Description	When active, allows open commands
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	4
Storage	Non-Volatile

8.5 Err_Latch

Classification	Fieldbus Device
Description	Specifies whether error indicators are to be retained in the Calc_Errrs, paramaters, etc.
FF Data Type	Unsigned8
Range	N/A
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

8.6 EXECUTION_TIME

Classification	Fieldbus Device
Description	The total time (in 1/32 milliseconds) required for the block algorithm to execute
FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	N/A

8.7 Expand_Dn

Classification	Fieldbus Device
-----------------------	-----------------

Description	Amount Lo and Lo_Lo are expanded after a change in the setpoint
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

8.8 Expand_Up

Classification	Fieldbus Device
Description	Amount Hi and Hi_Hi are expanded after a setpoint change
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

9. F

9.1 Fail

Classification	Fieldbus Device
Description	Enumeration of the current failure condition
FF Data Type	Unsigned16
Range	1 PassiveConfirmTime 2 State_1ConfirmTime 3 State_2ConfirmTime 4 PassiveConfirmLost 5 State_1ConfirmLost 6 State_2ConfirmLost 7 Tripped 8 LostInterlock 9 LockedOut 10 LostPower 11 MultipleConfirms 12 MultipleConfirms
Usage	Contained
Length	2
Valid View(s)	1,3
Storage	Dynamic

9.2 Fail_Alm

Classification	Fieldbus Device
Description	Generated when a confirm times out or is lost
FF Data Type	Alarm Discrete
Range	N/A
Usage	Contained
Length	13
Valid View(s)	N/A
Storage	Dynamic

9.3 Fail_Pri

Classification	Fieldbus Device
Description	Priority of the fail alarm
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

9.4 Fault_State

Classification	Fieldbus Device
Description	Condition set by loss of communication to an output block, failure promoted to an output block or a physical contact. When faultstate condition is set, then output function blocks will perform their FSTATE actions.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Clear 2 Active
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Non-Volatile

9.5 FB_CIDST

Classification	TPSB Only
Description	Function Block Control Input Destination. Defines the input parameter within the function block that is the destination for the value
FF Data Type	N/A
Range	N/A
Usage	N/A
Length	N/A
Valid View(s)	N/A

Storage	N/A
---------	-----

9.6 FB_CISRC

Classification	TPSB Only
Description	Function Block Control Input Source. Defines the parameter that provides the value to the function block. The parameter is normally an output parameter.
FF Data Type	N/A
Range	N/A
Usage	N/A
Length	N/A
Valid View(s)	N/A
Storage	N/A

9.7 FB_CITYPE

Classification	TPSB Only
Description	Function Block Control Input LinkageType. Linkage type may be one of the following: Not Configurer, Local, Client or Subscriber.
FF Data Type	N/A
Range	N/A
Usage	N/A
Length	N/A
Valid View(s)	N/A
Storage	N/A

9.8 FB_NoCIn

Classification	TPSB Only
Description	Defines the number of control inputs to a function block
FF Data Type	N/A
Range	N/A
Usage	N/A
Length	N/A
Valid View(s)	N/A
Storage	N/A

9.9 Feature_Sel

Classification	Fieldbus Device
Description	Used to select resource block options that are associated with the Features parameter.
FF Data Type	Bit String
Range	0 Unicode 1 Reports 2 Fault_State 3 Soft_W_Lock 4 Hard_W_Lock 5 Out_Readback 6 Direct_Write 7 ChangeBypassInAutoMode
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

9.10 Features

Classification	Fieldbus Device
Description	Used to show supported resource block options.
FF Data Type	Bit String
Range	0 Unicode 1 Reports 2 Fault_State 3 Soft_W_Lock 4 Hard_W_Lock 5 Out_Readback 6 Direct_Write 7 ChangeBypassInAutoMode
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

9.11 FF_Gain

Classification	Fieldbus Device
Description	The gain that the feed forward input is multiplied by before it is added to the calculated control output.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

9.12 FF_Scale

Classification	Fieldbus Device
Description	The feed forward input high and low scale values, engineering units code, and number of digits to the right of the decimal point.
FF Data Type	Scaling
Range	0-100
Usage	Contained
Length	11
Valid View(s)	4
Storage	Static

9.13 FF_Val

Classification	Fieldbus Device
Description	The feed forward value and status.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	3
Storage	Non-Volatile

9.14 Field_Val

Classification	Fieldbus Device
Description	Raw value of the field device in % of PV range, with a status reflecting the transducer condition, before signal characterization (L_TYPE) or filtering (PV_FTIME).
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Dynamic

9.15 Field_Val_D

Classification	Fieldbus Device
Description	Raw value of the field device discrete input, with a status reflecting the transducer condition.
FF Data Type	Value & Status – Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

9.16 FileDate

Classification	CFF File
Description	Capabilities file date or Value file date.
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

9.17 FileType

Classification	CFF File
Description	Defines the file type (capability or value)

FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

9.18 Final_Position_Value

Classification	Fieldbus Device
Description	The actual valve position and status. Could be used as the READBACK_VALUE in an AO block
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

9.19 Final_Value

Classification	Fieldbus Device
Description	The proposed valve position and status written by an analog output function block
FF Data Type	Value & Status Floating point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

9.20 Final_Value_Cutoff_Hi

Classification	Fieldbus Device
Description	Defines the cutoff high limit of the proposed value position
FF Data Type	Float
Range	Final_Value_Range, +INF

Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

9.21 Final_Value_Cutoff_Lo

Classification	Fieldbus Device
Description	Defines the cutoff low limit of the proposed value position
FF Data Type	Float
Range	Final_Value_Range, -INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

9.22 Final_Value_D

Classification	Fieldbus Device
Description	The proposed valve position and status written by a discrete function block
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

9.23 Final_Value_Range

Classification	Fieldbus Device
Description	The high and low range limit values, the engineering units code and the number of digits to the right of the decimal point to be used to display the final value
FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	1, 3
Storage	Static

9.24 Follow

Classification	Fieldbus Device
Description	If tracking is on, the output tracks the input
FF Data Type	Value & Status Discrete
Range	0 Off 1 Follow
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

9.25 Free_Space

Classification	Fieldbus Device
Description	Percent of memory available for further configuration. Zero in a preconfigured device.
FF Data Type	Float
Range	0-100%
Usage	Contained
Length	4
Valid View(s)	2
Storage	Dynamic

9.26 Free_Time

Classification	Fieldbus Device
Description	Percent of the block processing time that is free to process additional blocks.
FF Data Type	Float
Range	0-100%
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

9.27 FState_Time

Classification	Fieldbus Device
Description	Faultstate Time. The time in seconds from detection of failure of the output block remote setpoint to the output action of the block output if the condition still exists.
FF Data Type	Float
Range	positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

9.28 FState_Val

Classification	Fieldbus Device
Description	The preset analog SP value to use when failure occurs. This value will be used if the I/O option Faultstate-to-value is selected.
FF Data Type	Float
Range	PV_SCALE +/- 10%
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

9.29 FState_Val_D

Classification	Fieldbus Device
Description	The preset discrete (SP_D) value to use when failure occurs. This value will be used if the I/O option Faultstate-to-value is selected.
FF Data Type	Unsigned8
Range	N/A
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

10. G

10.1 Gain

Classification	Fieldbus Device
Description	Dimensionless value used by the block algorithm in calculating the block output.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

10.2 Gain_In_1 – Gain_In_3

Classification	Fieldbus Device
Description	Constant to be multiplied by (IN_n + bias)
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

10.3 Good_Lim

Classification	Fieldbus Device
Description	Sets the limit for percentage of inputs with good status (PCT_INCL)
FF Data Type	Float
Range	0-100
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

10.4 Grant_Deny

Classification	Fieldbus Device
Description	Options for controlling access of host computers and local control panels to operating, tuning, and alarm parameters of the block.
FF Data Type	Access Permissions
Range	0 Program 1 Tune 2 Alarm 3 Local
Usage	Contained
Length	2
Valid View(s)	2
Storage	Dynamic

11. H

11.1 Hard_Type

Classification	Fieldbus Device
Description	The types of hardware available as channel numbers.
FF Data Type	Bit String
Range	0 Scaler_Input 1 Scaler_Output 2 Discrete_Input 3 Discrete_Output
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

11.2 Hi_Alm

Classification	Fieldbus Device
Description	The status for high alarm and its associated time stamp.
FF Data Type	Alarm Float
Range	N/A
Usage	Contained
Length	16
Valid View(s)	N/A
Storage	Dynamic

11.3 Hi_Bias

Classification	Fieldbus Device
Description	This bias is added to $PSP \times Hi_Gain$ to get Hi_Lim
FF Data Type	Float
Range	Positive (Initial value is 0.0)
Usage	Contained
Length	4

Valid View(s)	2
Storage	Static

11.4 Hi_Gain

Classification	Fieldbus Device
Description	This gain multiplies the PSP (process actpoint) before addition of biases.
FF Data Type	Float
Range	(Initial value is 1.1)
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

11.5 Hi_Hi_Alm

Classification	Fieldbus Device
Description	The status for high high alarm and its associated time stamp.
FF Data Type	Alarm Float
Range	N/A
Usage	Contained
Length	16
Valid View(s)	N/A
Storage	Dynamic

11.6 Hi_Hi_Bias

Classification	Fieldbus Device
Description	The bias value added to $PSP \times Hi_Gain$ to get Hi_Hi_Gain
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

11.7 Hi_Hi_Lim

Classification	Fieldbus Device
Description	The value of the high high alarm trip setting in engineering units.
FF Data Type	Float
Range	PV_SCALE, +INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

11.8 Hi_Hi_Limx

Classification	Fieldbus Device
Description	The dynamic calculated Hi_Hi_Lim value.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

11.9 Hi_Limx

Classification	Fieldbus Device
Description	The dynamic calculated Hi_Lim
FF Data Type	Float
Range	
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

11.10 Hi_Hi_Pri

Classification	Fieldbus Device
-----------------------	-----------------

Description	Specifies the priority of the high high alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

11.11Hi_Lim

Classification	Fieldbus Device
Description	Specifies the value for the high alarm trip setting in engineering units.
FF Data Type	Float
Range	PV_SCALE, +INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

11.12Hi_Pri

Classification	Fieldbus Device
Description	Specifies the priority of the high alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

11.13Hysteresis

Classification	Fieldbus Device
Description	Defines the difference between the switching points.
FF Data Type	Float
Range	Positive

Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

12.1

12.1 Ignore

Classification	Fieldbus Device
Description	Specifies the inputs to be ignored for the device control function.
FF Data Type	Bit String
Range	0 Permissive 1 Interlock 2 Shutdown 3 PassiveConfirm 4 State1Confirm 5 State2Confirm
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

12.2 Ignore_Alm

Classification	Fieldbus Device
Description	Generated when a bit in the Ignore word is set
FF Data Type	Alarm Discrete
Range	N/A
Usage	Contained
Length	13
Valid View(s)	N/A
Storage	Dynamic

12.3 Ignore_Pri

Classification	Fieldbus Device
Description	Defines the priority of the Ignore alarm
FF Data Type	Unsigned8
Range	0-15
Usage	Contained

Length	1
Valid View(s)	4
Storage	Static

12.4 Ignore_Time

Classification	Fieldbus Device
Description	Defines time (in seconds) allowed to ignore new alarm
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

12.5 In

Classification	Fieldbus Device
Description	Primary input value of the block. Required for blocks that filter the input to get the PV.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	3
Storage	Non-Volatile

12.6 IN_1 – IN_4

Classification	Fieldbus Device
Description	Auxiliary input value to the block. Used for other values than the PV.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	3

Storage	Non-Volatile
----------------	--------------

12.7 In_Array

Classification	Fieldbus Device
Description	Array of input or X variables
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

12.8 In_D

Classification	Fieldbus Device
Description	Raw confirmation data
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	4
Storage	Non-Volatile

12.9 In_D1 – IN_D16

Classification	Fieldbus Device
Description	Discrete inputs for calculate block
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

12.10 In_Lo

Classification	Fieldbus Device
Description	Defines the value of the transmitter low range. This value is in the same engineering units as the PV.
FF Data Type	Value & Status Floating point
Range	N/A
Usage	Input
Length	5
Valid View(s)	4
Storage	Non-Volatile

12.11 Init_Val_1 – Init_Val_4

Classification	Fieldbus Device
Description	Initial value and status of the corresponding interface variable (IV_1 – IV_4)
FF Data Type	Value & Status Floating point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	4
Storage	Static

12.12 Init_Val_D1 – Init_Val_D4

Classification	Fieldbus Device
Description	Initial value and status of discrete inputs (IV_D1 – IV_D4)
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

12.13 Input_Opts

Classification	Fieldbus Device
Description	Defines the options that control the use of auxiliary inputs in the arithmetic.

FF Data Type	Bit String
Range	0 IN-UseUncertain 1 IN_LO-UseUncertain 2 IN_1-UseUncertain 3 IN_1-UseBad 4 IN_2-UseUncertain 5 IN_2-UseBad 6 INJ_3-UseUncertain 7 IN_3-UseBad
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

12.14Inst_Err_No

Classification	Fieldbus Device
Description	Displays the number of the instruction that sets an error in Calc_Err
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Dynamic

12.15Inst_Ex_Lim

Classification	Fieldbus Device
Description	Specifies the instruction execution count limit.
FF Data Type	Unsigned16
Range	Positive
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

12.16Inst_List1

Classification	Fieldbus Device
Description	Defines value of instruction list segment 1.
FF Data Type	Bit String
Range	N/A
Usage	Contained
Length	2
Valid View(s)	
Storage	Static

12.17Integ_Opts

Classification	Fieldbus Device
Description	Defines the configuration of the input type and flow direction in the totalizer function.
FF Data Type	Bit String
Range	0 Input1Accumulate 1 Input2Accumulate 2 FlowForward 3 FlowReverse 4 UseUncertain 5 UseBad 6 Carry
Usage	Contained
Length	2

Valid View(s)	4
Storage	Static

12.18Integ_Type

Classification	Fieldbus Device
Description	Determines the direction of counting and type of resetting in the integrator function.
FF Data Type	Unsigned8
Range	1 UP_AUTO 2 UP_DEM 3 DN_AUTO 4 DN_DEM 5 PERIODIC 6 DEMAND 7 PER&DEM
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

12.19Interlock_D

Classification	Fieldbus Device
Description	Input that duplicates shutdown
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Dynamic

12.20 Invert_Opts

Classification	Fieldbus Device
Description	If the bit is set, the corresponding input or output is inverted (Timer or Discrete Alarm function)
FF Data Type	Bit String
Range	0 InvertIN_D1 1 InvertIN_D2 2 InvertIN_D3 3 InvertIN_D4 4 InvertOUT_D 5 InvertOut_ALM
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

12.21 IO_Opts

Classification	Fieldbus Device
Description	Option which the user may select to alter input and output block processing.
FF Data Type	Bit String
Range	0 Invert 1 SP_tracks_PV_if_Man 2 Reserved 3 SP_tracks_PV_if_LO 4 SP_tracks_RCas_or_Cas_if_LO_or_Man 5 Increase_to_Close 6 FaultState_Type 7 FaultState_Restart 8 Target_to_Man 9 PV_for_BkCal_Out
Usage	Contained
Length	2
Valid View(s)	4

Storage	Static
----------------	--------

12.22IV_1 – IV_4

Classification	Fieldbus Device
Description	Interface variable
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

12.23IV_D1 – IV_D4

Classification	Fieldbus Device
Description	Interface variable
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

12.24IV_Desc_1 – IV_Desc_4

Classification	Fieldbus Device
Description	Description of interface variables IV_1 – IV_4.
FF Data Type	Octet String
Range	N/A
Usage	Contained
Length	64
Valid View(s)	
Storage	Static

12.25IV_Scale_1 – IV_Scale_4

Classification	Fieldbus Device
Description	High and low scale values for interface values IV_1 – IV_4.
FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	2
Storage	Static

12.26 IV_State_1 – IV_State_4

Classification	Fieldbus Device
Description	Index to the text describing the states discrete interface variables IV_D1-IV_D4.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

13. J

13.1 Jog_Time

Classification	Fieldbus Device
Description	Duration of OUT_D in active state (in seconds); jog open/closed
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

14. L

14.1 Lag_Time

Classification	Fieldbus Device
Description	Specifies the lag time constant (in seconds) for the block
FF Data Type	Value & Status Floating Point
Range	positive
Usage	Input
Length	5
Valid View(s)	4
Storage	Non-Volatile

14.2 Lead_Time

Classification	Fieldbus Device
Description	Specifies the lead time constant (in seconds) applied to input.
FF Data Type	Value & Status Floating Point
Range	positive
Usage	Input
Length	5
Valid View(s)	4
Storage	Non-Volatile

14.3 L_Type

Classification	Fieldbus Device
Description	Determines if the values passed by the transducer block to the AI block may be used directly (Direct) or if the value is in different units and must be converted linearly(Indirect) , or with square root (Ind Sqr Root), using the input range defined for the transducer and the associated output range.
FF Data Type	Unsigend8
Range	0 Uninitialized 1 Discrete 2 Indirect 3 Indirect_Sq_Root
Usage	Contained

Length	1
Valid View(s)	4
Storage	Static

14.4 Level_Offset

Classification	Fieldbus Device
Description	The distance from the empty calibration level to a reference point, such as the bottom of the tank.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Non-Volatile

14.5 Lim_Notify

Classification	Fieldbus Device
Description	Defines the maximum number of unconfirmed alert notify messages allowed.
FF Data Type	Unsigned8
Range	0-Max_Notify
Usage	Contained
Length	1
Valid View(s)	2
Storage	Static

14.6 __Link_Object

Classification	Fieldbus Device
Description	This parameter contains a null set of values that will clear the Link_Object(s) in a device. This clear value is sent to the device at the start of configuration down load.
FF Data Type	FB Link
Range	N/A
Usage	N/A
Length	8

Valid View(s)	N/A
Storage	Non Volatile

14.7 Link_Object

Classification	Fieldbus Device
Description	Specifies the function block VCR linkage and VCR OD Index and Service Operation for a block parameter. This parameter is automatically built during the creation of the LAS Schedule.
FF Data Type	FB Link
Range	N/A
Usage	N/A
Length	8
Valid View(s)	N/A
Storage	Non Volatile

14.8 LinkScheduleActivationVariable

Classification	Fieldbus Device
Description	Link schedule activation variable.
FF Data Type	Unsigned16
Range	N/A
Usage	N/A
Length	2
Valid View(s)	N/A
Storage	N/A

14.9 Lin_Type

Classification	Fieldbus Device
Description	The linearization type used to describe the behavior of the sensor output.
FF Data Type	Unsigned8
Range	0 Undefined 1 Linear-Input 2 Linear-Output 3 SquareRoot

	4 SquareRootThirdPower
	5 SquareRootFifthPower
	255 Other
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

14.10Lo_Alm

Classification	Fieldbus Device
Description	The status of the low alarm and its associated time stamp.
FF Data Type	Alarm Float
Range	N/A
Usage	Contained
Length	16
Valid View(s)	N/A
Storage	Dynamic

14.11Lo_Bias

Classification	Fieldbus Device
Description	This bias is subtracted from PSP Lo_Gain→Lo_Lim.
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

14.12Lo_Gain

Classification	Fieldbus Device
Description	This gain multiplies PSP (____ setpoint) before subtraction of biases
FF Data Type	Float
Range	N/A

Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

14.13Lo_Lim

Classification	Fieldbus Device
Description	Defines the trip setting for the low alarm in engineering units.
FF Data Type	Float
Range	-INF, PV_SCALE
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

14.14Lo_Limx

Classification	Fieldbus Device
Description	The dynamic calculated Lo_Lim.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

14.15Lo_Lo_Alm

Classification	Fieldbus Device
Description	The status of the low low alarm and its associated time
FF Data Type	Alarm Float
Range	N/A
Usage	Contained
Length	16
Valid View(s)	N/A
Storage	Dynamic

14.16Lo_Lo_Bias

Classification	Fieldbus Device
Description	Specifies the bias is subtracted from $PSP * Lo_Gain - >Lo_LoLim$
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

14.17Lo_Lo_Lim

Classification	Fieldbus Device
Description	Specifies the trip setting of the low low alarm in engineering units.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

14.18Lo_Lo_Limx

Classification	Fieldbus Device
Description	The dynamic calculated Lo_Lo_Lim.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

14.19Lo_Lo_Pri

Classification	Fieldbus Device
Description	Defines the priority of the low low alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

14.20Lo_Pri

Classification	Fieldbus Device
Description	Defines the priority of the low alarm.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

14.21 Lockval

Classification	Fieldbus Device
Description	Defines the flag for holding the first output at the current value.
FF Data Type	Unsigned8
Range	1 NoLock 2 Lock
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

14.22 Low_Cut

Classification	Fieldbus Device
Description	Defines the limit used in square root processing. A value of zero percent-of-scale is used in block processing if the transducer value falls below this limit, in % of scale. This feature may be used to eliminate noise near zero from a flow
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

15. M

15.1 Manufac_ID

Classification	Fieldbus Device
Description	Manufacturer identification number - used by an interface device to locate the DD file for the resource.
FF Data Type	Unsigned32
Range	N/A
Usage	N/A
Length	2
Valid View(s)	N/A
Storage	N/A

15.2 Max_Notify

Classification	Fieldbus Device
Description	Defines the maximum number of possible unconfirmed alert notify messages possible.
FF Data Type	Unsigned8
Range	N/A
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

15.3 Memory_Size

Classification	Fieldbus Device
Description	Defines the available configuration memory in the empty resource. This value is to be checked before attempting a download.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

15.4 Min_Cycle_T

Classification	Fieldbus Device
Description	Time duration of the shortest cycle interval of which the resource is capable.
FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

15.5 Min_Good

Classification	Fieldbus Device
Description	Shows the minimum number of good inputs
FF Data Type	Unsigned8
Range	0-4
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

15.6 Mode_Blkl

Classification	Fieldbus Device
Description	The actual, target, permitted, and normal modes of the block.
FF Data Type	4 Bit Strings
Range	0 ROut 1 RCas 2 Cas 3 Auto 4 Man 5 LO 6 IMan 7 OOS

Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Mixed

15.7 Mode_Blк_AI_Set

Classification	TPSB Only
Description	Used by TPSB after configuration download to set the target mode for the AI Block.
FF Data Type	4 Bit Strings
Range	3 Auto 4 Man 7 OOS
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	Mixed

15.8 Mode_Blк_AO_Set

Classification	TPSB Only
Description	Used by TPSB after configuration download to set the target mode for the AO Block.
FF Data Type	4 Bit Strings
Range	1 RCas 2 Cas 3 Auto 4 Man 7 OOS
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	Mixed

15.9 Mode_Blк_PID_Set

M**Mode_Blк_Resource_Set**

Classification	TPSB Only
Description	Used by TPSB after configuration download to set the target mode for the PID Block.
FF Data Type	4 Bit Strings
Range	0 ROut 1 RCas 2 Cas 3 Auto 4 Man 7 OOS
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	Mixed

15.10Mode_Blк_Resource_Set

Classification	TPSB Only
Description	Used by TPSB after configuration download to set the target mode for the ResourceBlock.
FF Data Type	4 Bit Strings
Range	3 Auto 7 OOS
Usage	Contained
Length	4
Valid View(s)	N/A
Storage	Mixed

15.11Mode_Blк_AI_OOS

Classification	Fieldbus Device
Description	Used by the TPSB during configuration down load to set the AI block Out of Service.
FF Data Type	4 Bit Strings
Range	7 OOS
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Mixed

15.12Mode_Blк_AO_OOS

Classification	Fieldbus Device
Description	Used by the TPSB during configuration down load to set the AO block Out of Service.
FF Data Type	4 Bit Strings
Range	7 OOS
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Mixed

15.13Mode_Blк_PID_OOS

Classification	Fieldbus Device
Description	Used by the TPSB during configuration down load to set the PID block Out of Service.
FF Data Type	4 Bit Strings
Range	7 OOS
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Mixed

15.14Mode_Blк_Resource_OOS

Classification	Fieldbus Device
Description	Used by the TPSB during configuration down load to set the Resource block Out of Service.
FF Data Type	4 Bit Strings
Range	7 OOS
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Mixed

15.15ModelName

Classification	Fieldbus Device
Description	The model name of the instrument. Resides in the VFD
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

15.16Module_Sn

Classification	Fieldbus Device
Description	The serial number of the electronics module in the instrument.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	4
Storage	Static

16. N

16.1 Next_FB_To_Execute

Classification	Fieldbus Device
Description	Specifies the object dictionary index for the block object of a block which will be invoked to execute following the completion of execution of this block.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A
Storage	N/A

16.2 N_Reset

Classification	Fieldbus Device
Description	Displays the number of resets
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

16.3 N_Start

Classification	Fieldbus Device
Description	Displays the number of false-to-true transitions (combined input).
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Dynamic

16.4 NodeAddress

Classification	Fieldbus Device
Description	Defines the node address of the fieldbus device on the link
FF Data Type	Unsigned16
Range	N/A
Usage	N/A
Length	2
Valid View(s)	N/A
Storage	N/A

16.5 Num_Of_Params

Classification	Fieldbus Device
Description	The number of parameters associated with the function block including the block header parameter.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A
Storage	N/A

16.6 Number_View_3 – Number_View_4

Classification	Fieldbus Device
Description	Number of entries in the object dictionary for View_3 and View_4 support, respectively.
FF Data Type	Unsigned8
Range	N/A
Usage	Contained
Length	1
Valid View(s)	N/A
Storage	N/A

16.7 NV_Cycle_T

Classification	Fieldbus Device
Description	Interval between writing copies of non-volatile parameters to non-volatile memory. Zero means never.
FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

17.0

17.1 Op_Cmd_CXO

Classification	Fieldbus Device
Description	This command sets the discrete point into the setpoint state.
FF Data Type	Bit String
Range	0 Reserved 1 Close 2 Open 3 Stop 4 Enable-1-2-3
Usage	Contained
Length	2
Valid View(s)	3
Storage	Dynamic

17.2 Op_Cmd_Int

Classification	Fieldbus Device
Description	This command rests the integrator.
FF Data Type	Unsigned8
Range	0 OFF 1 RESET
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Dynamic

17.3 Op_Cmd_Spg

Classification	Fieldbus Device
Description	This operator command enables the positions in a profile in the setpoint ramp.
FF Data Type	Unsigned8
Range	0 UNDEFINED

	1 RESET_IN
	2 ADVANCED
	3 REPEAT
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Dynamic

17.4 Op_Select

Classification	Fieldbus Device
Description	Specifies an operator settable param to force a given input
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

17.5 Out

Classification	Fieldbus Device
Description	The primary analog value calculated as a result of executing the function block.
FF Data Type	Value & Status Floating Point
Range	Out_Scale +/- 10%
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

17.6 Out_1 – Out_2

Classification	Fieldbus Device
Description	Defines the block outputs required by splitter.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

17.7 Out_1_State – Out_4_State

Classification	Fieldbus Device
Description	Units code for discrete output 1
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

17.8 Out_1_Units – Out_4_Units

Classification	Fieldbus Device
Description	Defines the engineering units code for discrete outputs 1-4.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

17.9 Out_Accum

Classification	Fieldbus Device
Description	Specifies the number of accumulated counts.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

17.10 Out_Alm

Classification	Fieldbus Device
Description	The alarm summary variable of the analog block
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Output
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

17.11 Out_Alm_Sum

Classification	Fieldbus Device
Description	Specifies the alarm conditions for a true Out_Alm in the analog alarm.
FF Data Type	Unsigned8
Range	N/A
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

17.12 Out_Array

Classification	Fieldbus Device
Description	Array of output or Y variables
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

17.13 Out_D

Classification	Fieldbus Device
Description	The primary discrete value calculated as a result of executing the function block.
FF Data Type	Value & Status Discrete
Range	Out_State
Usage	Output
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

17.14 Out_D1 – Out_D4

Classification	Fieldbus Device
Description	The calculated discrete output value for this block
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Output
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

17.15Out_Exp

Classification	Fieldbus Device
Description	Specifies the time expired. Stops at Timer_SP
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

17.16Out_Hi_Lim

Classification	Fieldbus Device
Description	Defines the maximum output hi limit value for modes other than Manual.
FF Data Type	Float
Range	OUT_SCALE +/- 10%
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

17.17Out_Lo_Lim

Classification	Fieldbus Device
Description	Defines the minimum output limit value for modes other than Manual.
FF Data Type	Float
Range	OUT_SCALE +/- 10%
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

17.18 Out_Ptrip

Classification	Fieldbus Device
Description	Defines the state and value of the second discrete output.
FF Data Type	Value & Status Discrete
Range	0 OFF 1 ON
Usage	Output
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

17.19 Out_Rem

Classification	Fieldbus Device
Description	Specifies the time remaining if the timer is active
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

17.20 Out_Scale

Classification	Fieldbus Device
Description	Defines the high and low scale values, engineering units code, and number of digits to the right of the decimal point to be used in displaying the Out parameter and parameters which have the same scaling as Out.
FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	2
Storage	Static

17.21Out_State

Classification	Fieldbus Device
Description	Specifies the index number to the text describing the state of a discrete output.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

17.22Out_Trip

Classification	Fieldbus Device
Description	Specifies the alarm trip state of the first discrete output
FF Data Type	Value & Status Discrete
Range	0 Off 1 On
Usage	Output
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

17.23Out_Units

Classification	Fieldbus Device
Description	Defines the engineering units that are to be displayed for the output.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

17.24Outage_Lim

Classification	Fieldbus Device
Description	Defines the maximum tolerated duration for power failure
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

18. P

18.1 Pause

Classification	Fieldbus Device
Description	This command stops the timer when the command is set to true
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

18.2 Pause_Cause

Classification	Fieldbus Device
Description	Specifies the cause of a pause.
FF Data Type	Unsigned8
Range	0 NotPaused 1 OperatorPause 2 LogicPause 3 DeviationPause 4 Operator&Logic 5 Operator&Deviation 6 Logic&Deviation 7 Operator&Logic&Deviation
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Dynamic

18.3 Pct_Incl

Classification	Fieldbus Device
Description	Displays the percentage of inputs with good static's
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Dynamic

18.4 PDTag

Classification	Fieldbus Device
Description	Defines the physical device tag
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

18.5 Permissive_D

Classification	Fieldbus Device
Description	Specifies the input that prevents Out_D from going to the active state
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

18.6 Pre_Out

Classification	Fieldbus Device
Description	Displays the output value and status.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Dynamic

18.7 Pre_Out_1 – Pre_Out_2

Classification	Fieldbus Device
Description	Displays output value and status of pre-output 1 and 2
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Dynamic

18.8 Pre_Out_Alm

Classification	Fieldbus Device
Description	Specifies the alarm summary variable of an analog alarm block
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

18.9 Pre_Out_D

Classification	Fieldbus Device
Description	Specifies the alarm summary variable of a discrete.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

18.10Pre_Out_D1 – Pre_Out_D4

Classification	Fieldbus Device
Description	Specifies the alarm summary variable calculated discrete variables 1-4
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

18.11Pre_Trip

Classification	Fieldbus Device
Description	Adjusts the quantity that will set Out_Ptrip
FF Data Type	Float
Range	positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

18.12Primary_Value

Classification	Fieldbus Device
Description	The measured value and status available to the function block.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Dynamic

18.13Primary_Value_Range

Classification	Fieldbus Device
Description	The High and Low range limit values, the engineering units code, and the number of digits to the right of the decimal point to be used to display the primary value.
FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	2, 4
Storage	Static

18.14Primary_Value_Type

Classification	Fieldbus Device
Description	The type of measurement represented by the primary value. For example: Mass Flow, Absolute Pressure, Differential Temperature.
FF Data Type	Unsigned16
Range	1 Other 100 MassFlow 101 VolumetricFlow 102 AverageMassFlow 103 AverageVolumetricFlow 104 ProcessTemperature

	105 Non-ProcessTemperature
	106 DifferentialTemperature
	107 DifferentialPressure
	108 GaugePressure
	109 AbsolutePressure
	110 Level
	111 pH (<i>picohenry</i>)
	112 mV (<i>millivolt</i>)
	113 Conductivity
Usage	Contained
Length	2
Valid View(s)	2, 4
Storage	Static

18.15PROFILE

Classification	Fieldbus Device
Description	Number assigned by the FOUNDATION Fieldbus which uniquely identifies the profile on which the block definition is based
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A
Storage	N/A

18.16Profile_Revision

Classification	Fieldbus Device
Description	The revision number of the profile on which the block definition is based.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A

Storage	N/A
----------------	-----

18.17Prop_Lims

Classification	Fieldbus Device
Description	Defines whether the limits are propagated to the result
FF Data Type	Unsigned8
Range	0 = False 1 = True
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

18.18PSP

Classification	Fieldbus Device
Description	Specifies the value of the process setpoint
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

18.19Pulse_Dur

Classification	Fieldbus Device
Description	Specifies the pulsewidth, in seconds, of the integral pulse
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

18.20Pulse_Val and Pulse_Val1 – Pulse_Val2

Classification	Fieldbus Device
Description	The value of each metered pulse in engineering units.
FF Data Type	Float
Range	positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

18.21PV

Classification	Fieldbus Device
Description	The primary analog value for use in executing the function, or a process value associated with the function. May also be calculated from the READBACK value of an AO block.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Dynamic

18.22PV_D

Classification	Fieldbus Device
Description	Defines the primary discrete value for use in executing the function, or a process value associated with it. May also be calculated from the Readback_D parameter value of a digital output.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

18.23PV_Ftimex

Classification	Fieldbus Device
Description	Specifies the time constant (in seconds) of a single exponential filter for the PV.
FF Data Type	Float
Range	positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

18.24PV_Scale

Classification	Fieldbus Device
Description	Defines the high and low scale values, engineering units code, and number of digits to the right of the decimal point to be used in displaying the PV parameter and parameters which have the same scaling as PV.
FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	2
Storage	Static

18.25PV_State

Classification	Fieldbus Device
Description	This parameter contains the index number of the text describing the state of a discrete PV within the object dictionary.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

18.26PV_Units

Classification	Fieldbus Device
Description	Defines the engineering units index for the units that are to be displayed.
FF Data Type	Unsigned16
Range	See Units
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

19. Q

19.1 Quies_Opt

Classification	Fieldbus Device
Description	Options for Out_Exp and Out_Rem during quiescence
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Clear 2 Last
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

20. R

20.1 RA_Ftime

Classification	Fieldbus Device
Description	Defines the time constant (in seconds) of a single exponential filter for the value to be ratioed..
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

20.2 Range_Hi

Classification	Fieldbus Device
Description	Defines the value of the constant used in switching to HI Range.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

20.3 Range_Lo

Classification	Fieldbus Device
Description	Defines the value of the constant used in switching to LO Range.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

20.4 Rate

Classification	Fieldbus Device
Description	Defines the derivative time constant in seconds.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

20.5 RCas_In

Classification	Fieldbus Device
Description	The target setpoint and status provided by a supervisory host to a analog control or output block.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	3
Storage	Non-Volatile

20.6 RCas_In_D

Classification	Fieldbus Device
Description	The target setpoint and status provided by a supervisory host to a discrete control or output block.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	3
Storage	Non-Volatile

20.7 RCas_Out

Classification	Fieldbus Device
-----------------------	-----------------

Description	The block setpoint and status after ramping - provided to a supervisory host for back calculation and to allow action to be taken under limiting conditions or mode change.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	3
Storage	Dynamic

20.8 RCas_Out_D

Classification	Fieldbus Device
Description	The block setpoint and status provided to a supervisory host for back calculation and to allow action to be taken under limiting conditions or mode change.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	3
Storage	Dynamic

20.9 ReadBack

Classification	Fieldbus Device
Description	Indicates the readback value of the actual continuous valve or other actuator position, in transducer units.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Dynamic

20.10 ReadBack_D

Classification	Fieldbus Device
Description	Indicates the readback state of the actual discrete valve or

	other actuator position, in the transducer state.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Dynamic

20.11Release

Classification	Fieldbus Device
Description	Manufacturer release information. Resides in the VFD
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

20.12Reset

Classification	Fieldbus Device
Description	Defines the integral time constant (in seconds) per repeat.
FF Data Type	Float
Range	positive, +INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

20.13Reset_Confirm

Classification	Fieldbus Device
Description	Momentary discrete value which must be written by a host device to enable further resets.
FF Data Type	Unsigned8
Range	0 Off 1 Confirm
Usage	Contained
Length	1
Valid View(s)	3
Storage	Dynamic

20.14Reset_D

Classification	Fieldbus Device
Description	Momentary switch that clears the locked out condition or retries the faile state.
FF Data Type	Unsigned8
Range	0 Off 1 Reset
Usage	Contained
Length	1
Valid View(s)	3
Storage	Dynamic

20.15Reset_In

Classification	Fieldbus Device
Description	This command resets the timer
FF Data Type	Value & Status Discrete
Range	0 Off 1 Reset
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

20.16Restart

Classification	Fieldbus Device
Description	Allows a manual restart to be initiated. Several degrees of restart are possible..
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Run 2 Restart Resource 3 Restart with Defaults 4 Restart Processor
Usage	Contained
Length	1
Valid View(s)	N/A
Storage	Dynamic

20.17Restart_Time

Classification	Fieldbus Device
Description	Specifies the maximum time required in the passive state.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

20.18Rev_Flow1 – Rev_Flow2

Classification	Fieldbus Device
Description	Indicates reverse flow when true
FF Data Type	Value & Status Discrete
Range	0 FORWARD 1 REVERSE
Usage	Input
Length	2
Valid View(s)	1, 3

Storage	Non-Volatile
---------	--------------

20.19ROut_In

Classification	Fieldbus Device
Description	The target output and status provided by a host to the control block for use as the output (ROut mode).
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	3
Storage	Non-Volatile

20.20ROut_Out

Classification	Fieldbus Device
Description	The block output and status provided to a host for back calculation in ROut mode and to allow action to be taken under limited conditions or mode change.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	3
Storage	Dynamic

20.21RS_State

Classification	Fieldbus Device
Description	The current state of the function block application-state machine.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 StartRestart 2 Initialization 3 Online_Linking 4 Online

	6 Failure
	5 Standby
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Dynamic

20.22Rtotal

Classification	Fieldbus Device
Description	Specifies the total number non-good statuses.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Non-Volatile

21. S

21.1 Safeguard_CI

Classification	Fieldbus Device
Description	Defines the high-priority discrete setpoint
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	4
Storage	Non-Volatile

21.2 Safeguard_Op

Classification	Fieldbus Device
Description	Defines the high-priority discrete output.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	4
Storage	Non-Volatile

21.3 SchName

Classification	TPSB Only
Description	Defines the strategy name for the LAS Schedule.
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

21.4 Secondary_Value

Classification	Fieldbus Device
Description	The secondary value related to the sensor.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Dynamic

21.5 Secondary_Value_Unit

Classification	Fieldbus Device
Description	The engineering units to be used with Secondary_Value parameter value.
FF Data Type	Unsigned16
Range	(see Appendix for UNITS values)
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

21.6 Sel_1 – Sel_3

Classification	Fieldbus Device
Description	Input value to the respective selector input (1-3).
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

21.7 Sel_Type

Classification	Fieldbus Device
Description	This parameter specifies the type of selector action.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 High 2 Low 3 Middle
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

21.8 Select_Type

Classification	Fieldbus Device
Description	This parameter specifies the selector action type.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 FirstGood 2 Min 3 Max 4 Mid 5 Avg
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

21.9 Selected

Classification	Fieldbus Device
Description	Integer indicating which input is selected
FF Data Type	Value & Status Discrete
Range	N/A

Usage	Output
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

21.10Sensor_Cal_Date

Classification	Fieldbus Device
Description	The date of the last sensor calibration.
FF Data Type	Date
Range	N/A
Usage	Contained
Length	7
Valid View(s)	4
Storage	Static

21.11Sensor_Cal_Loc

Classification	Fieldbus Device
Description	The last physical location at which the sensor was calibrated. (ex. Acme Labs)
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	4
Storage	Static

21.12Sensor_Cal_Method

Classification	Fieldbus Device
Description	The method of last sensor calibration. It could be one of the several standard calibration methods defined by ISO or some other method.
FF Data Type	Unsigned8
Range	100 Volumetric 101 StaticWeight 102 DynamicWeight

S**Sensor_Cal_Who**

	103 FactorTrimStandard
	104 UserTrimStandard
	105 FactoryTrimSpecial
	106 UserTrimSpecial
	255 Other
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

21.13Sensor_Cal_Who

Classification	Fieldbus Device
Description	The name of the person responsible for the last sensor calibration.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	N/A
Storage	Static

21.14Sensor_Connection

Classification	Fieldbus Device
Description	The number of wires for the temperature probe.
FF Data Type	Unsigned8
Range	2 2_wire
	3 3_wire
	4 4_wire
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

21.15Sensor_Fill_Fluid

Classification	Fieldbus Device
Description	Defines the type of fluid used in the sensor.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

21.16Sensor_Isolator_Mtl

Classification	Fieldbus Device
Description	Defines the construction material of the isolator
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

21.17Sensor_Offset

Classification	Fieldbus Device
Description	The distance from the sensor reference point to the top of the tank.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

21.18Sensor_Range

Classification	Fieldbus Device
Description	The High and Low range limit values, the engineering units code and the number of digits to the right of the decimal point for the sensor.

S**Sensor_Sn**

FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	4
Storage	Static

21.19 Sensor_Sn

Classification	Fieldbus Device
Description	The sensor serial number.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

21.20 Sensor_Type

Classification	Fieldbus Device
Description	Defines the sensor type.
FF Data Type	Unsigned16
Range	-1 Non-Standard
	100 FlowSensorUnknown
	101 Coriolis(Gyroscopic)
	102 Electromagnetic
	103 mV (<i>millivolt</i>)
	105 DeltaOhm
	104 Ohm
	106 NuclearMagneticResonance
	107 PositiveDisplacement
	108 Refraction
	109 Tagging
	110 Ultrasonic(Doppler)
	111 Ultrasonic(TimeOfTravel)

112 Vortex
 113 Target
 114 VariableArea
 115 LevelSensorUnknown
 116 Radar
 117 Capacitance
 118 Nuclear
 119 Ultrasonic
 120 FloatGauge
 121 PressureSensorUnknown
 122 ResonantWire
 123 VibrationBeam
 124 StrainGague
 125 PiezoResistive
 126 SiliconResonant
 127 TemperatureSensorUnknown
 128 PT100_A_385
 129 PT100_A_392
 130 PT200_A_385
 131 PT500_A_385
 132 NI120
 133 CU10
 134 T/C-Type_B
 135 T/C-Type_C
 136 T/C-Type_E
 137 T/C-Type_J
 138 T/C-Type_K
 139 T/C-Type_N
 140 T/C-Type_R
 141 T/C-Type_S
 142 T/C-Type_T
 143 T/C-Type-DIN_L
 144 T/C-Type-DIN_U

Usage

Contained

S**SequenceName**

Length	2
Valid View(s)	4
Storage	Static

21.21SequenceName

Classification	TPSB Only
Description	Defines the strategy name for a sequence in the LAS Schedule
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

21.22Servo_Gain

Classification	Fieldbus Device
Description	Defines the servo PID gain value.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

21.23 Servo_Rate

Classification	Fieldbus Device
Description	Defines the servo PID rate value.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

21.24 Servo_Reset

Classification	Fieldbus Device
Description	Defines the servo PID reset value.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

21.25 Set_Fstate

Classification	Fieldbus Device
Description	Allows the failsafe condition to be manually initiated by selecting Set.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Off 2 Set
Usage	Contained
Length	1
Valid View(s)	N/A
Storage	Dynamic

21.26 Shed_Opt

Classification	Fieldbus Device
Description	Defines the shed option for the remote control device.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 NormalShed_NormalReturn 2 NormalShed_NoReturn 3 ShedToAuto_NormalReturn 4 ShedToAuto_NoReturn 5 ShedToManual_NormalReturn 6 ShedToManual_NoReturn 7 ShedToRetainTarget_NormalReturn 8 ShedToRetainTarget_NoReturn
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

21.27Shed_Rcas

Classification	Fieldbus Device
Description	Function block will shed from Rcas to the specified shed option if a write command has not been received within this time duration.
FF Data Type	Unsigned32
Range	N/A
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

21.28Shed_Rout

Classification	Fieldbus Device
Description	Function block will shed from Rout to the specified shed option if a write command has not been received within this time duration.
FF Data Type	Unsigned32

Range	N/A
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

21.29Shutdown_D

Classification	Fieldbus Device
Description	Specified the input that causes Out_D to go to the passive state
FF Data Type	Value & Status Discrete
Range	0 Off 1 On
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

21.30Simulate

Classification	Fieldbus Device
Description	Allows manually-entered value for transducer analog I/O.
FF Data Type	Simulate - Floating Point
Range	N/A
Usage	Contained
Length	11
Valid View(s)	N/A
Storage	Dynamic

21.31Simulate_D

Classification	Fieldbus Device
Description	Allows manually-entered value for transducer discrete I/O.
FF Data Type	Simulate - Discrete
Range	N/A
Usage	Contained

Length	5
Valid View(s)	N/A
Storage	Dynamic

21.32 Simulate_P

Classification	Fieldbus Device
Description	Allows manually-entered value for transducer.
FF Data Type	Simulate - Floating Point
Range	N/A
Usage	Contained
Length	11
Valid View(s)	N/A
Storage	Dynamic

21.33 Simulate_S

Classification	Fieldbus Device
Description	Allows manually-entered value for transducer strings (PSEUDO)
FF Data Type	Simulate - Discrete
Range	N/A
Usage	Contained
Length	5
Valid View(s)	N/A
Storage	Dynamic

21.34 SP

Classification	Fieldbus Device
Description	The analog setpoint value of this block.
FF Data Type	Value & Status Floating Point
Range	PV_SCALE +/- 10%
Usage	Contained
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

21.35SP_D

Classification	Fieldbus Device
Description	The discrete setpoint value of this block.
FF Data Type	Value & Status Discrete
Range	PV_State
Usage	Contained
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

21.36Spg_State

Classification	Fieldbus Device
Description	Defines the operating state of the block in the setpoint ramp function
FF Data Type	Unsigned8
Range	0 Undefined 1 Ready 2 Active 3 Pause 4 AT_End
Usage	Contained
Length	1
Valid View(s)	1, 3
Storage	Dynamic

21.37SP_Hi_Lim

Classification	Fieldbus Device
Description	Defines the setpoint high limit. The setpoint high limit is the highest setpoint operator entry that can be used for the block.
FF Data Type	Float
Range	PV_SCALE +/- 10%
Usage	Contained
Length	4
Valid View(s)	2

Storage	Static
---------	--------

21.38SP_Lo_Lim

Classification	Fieldbus Device
Description	Defines the setpoint low limit. The setpoint low limit is the lowest setpoint operator entry that can be used for the block.
FF Data Type	Float
Range	PV_SCALE +/- 10%
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

21.39SP_Rate_Dn

Classification	Fieldbus Device
Description	Defines the ramp rate at which downward setpoint changes are acted on in Auto mode, in PV units per second. If the ramp rate is set to zero or the block is in a mode other than Auto, then the setpoint will be used immediately.
FF Data Type	Float
Range	Positive, +INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

21.40SP_Rate_Up

Classification	Fieldbus Device
Description	Ramp rate at which upward setpoint changes are acted on in Auto mode, in PV units per second. If the ramp rate is set to zero or the block is in a mode other than Auto, then the setpoint will be used immediately.
FF Data Type	Float
Range	positive, +INF
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

21.41SRTotal

Classification	Fieldbus Device
Description	This parameter contains the snapshot of the Rtotal value just before reset
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Non-Volatile

21.42SSP

Classification	Fieldbus Device
Description	The parameter contains the snapshot value of the setpoint.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Non-Volatile

21.43ST_Rev

Classification	Fieldbus Device
Description	This parameter contains the revision level of the static data associated with the function block. The revision value will be incremented each time a static parameter value in the block is changed.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	1, 2, 3, 4
Storage	Static

21.44 Start

Classification	Fieldbus Device
Description	A leading edge at this input
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

21.45 Start_Type

Classification	Fieldbus Device
Description	Specifies the starting point option in the setpoint ramp function.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 UseCurve 2 UseDuration 3 UseRate
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

21.46Start_Val

Classification	Fieldbus Device
Description	Specified the starting point of each segment of the setpoint profile
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	2
Storage	Static

21.47Status_Opts

Classification	Fieldbus Device
Description	Options which the user may select in the block processing of status. Status options are applicable to the following blocks: BG, PD, AI, DI, AO, ML, CS, RA, PID, and DO
FF Data Type	Bit String
Range	0 IFS_if_Bad_IN 1 IFS_if_Bad_Cas_IN 2 Uncertain_as_Good 3 Propagate_Fault_Fwd 4 Propagate_Fault_Bkwd 5 Target_to_Man_if_Bad_IN 6 Uncertain_if_Limited 7 Bad_if_Limited 9 Reserved
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

21.48Step_Posn

Classification	Fieldbus Device
Description	This parameter specifies the current step
FF Data Type	Value & Status Discrete

Range	N/A
Usage	Output
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

21.49 Stotal

Classification	Fieldbus Device
Description	This parameter contains a snapshot of the total value before a reset occurs.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Non-Volatile

21.50 Strategy

Classification	Fieldbus Device
Description	The strategy field can be used to identify grouping of blocks.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

21.51SubSchName

Classification	TPSB Only
Description	Defines the strategy name of a sub-schedule in the LAS Schedule.
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

21.52Swap_2

Classification	Fieldbus Device
Description	This parameter swaps In_2 to Y and Out_2 to X
FF Data Type	Unsigned8
Range	1 NoSwap 2 Swap
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

22. T

22.1 Tag_Desc

Classification	Fieldbus Device
Description	The user description of the block
FF Data Type	Octet String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	N/A
Storage	Static

22.2 Temp_1 – Temp_4

Classification	Fieldbus Device
Description	The parameter defines the temperatur setpoints.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Contained
Length	5
Valid View(s)	4
Storage	Dynamic

22.3 Temp_D1 – Temp_D8

Classification	Fieldbus Device
Description	Defines the discrete variable for temperature
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Dynamic

22.4 Test_RW

Classification	Fieldbus Device
Description	Specifies the read/write test parameter.
FF Data Type	Test
Range	N/A
Usage	Contained
Length	112
Valid View(s)	N/A
Storage	Dynamic

22.5 Time_Posn

Classification	Fieldbus Device
Description	This parameter determines the current step
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

22.6 Time_Posn_T

Classification	Fieldbus Device
Description	This parameter determines the current step
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Output
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

22.7 Time_Unit1 – Time_Unit2

Classification	Fieldbus Device
Description	These parameters specify the units of time for display in the pulse input, setpoint ramp, and timer functions.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Seconds 2 Hours 3 Minutes 4 Days 5 [day-[hr:[min[:sec]]]]
Usage	Contained
Length	1
Valid View(s)	2
Storage	Static

22.8 Time_Units

Classification	Fieldbus Device
Description	This parameter displays the time units for Timer_SP, Out_Exp, & Out_Rem
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Seconds 2 Hours 3 Minutes 4 Days 5 [day-[hr:[min[:sec]]]]
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

22.9 Total_Sp

Classification	Fieldbus Device
Description	The SP for batch totalization
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	1, 3
Storage	Static

22.10Timer_Type

Classification	Fieldbus Device
Description	Defines the type of time processing applied to the PV in the timer function.
FF Data Type	Unsigned8
Range	0 Measure 1 Accum 2 Compare 3 Delay 4 Extend 5 Debounce 6 Pulse 7 RT_Pulse
Usage	Contained
Length	1
Valid View(s)	4
Storage	Static

22.11Travel_Time

Classification	Fieldbus Device
Description	Specifies the required time for the actuator to move final control from a to b
FF Data Type	Float
Range	Positive

Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

22.12Travel_Timer

Classification	Fieldbus Device
Description	Displays the time taken to confirm a new state
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Dynamic

22.13Transducer_Directory

Classification	Fieldbus Device
Description	A directory that specifies the number and starting indices of the data collections in the transducer block.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A
Storage	Non-Volatile

22.14Transducer_Type

Classification	Fieldbus Device
Description	Identifies the transducer type as shown in the following listing.
FF Data Type	Unsigned16
Range	1 Other 100 Pressure-Calibrated 101 Temperature-Calibrated

	102 DualTemperature-Calibrated
	103 RadarLevel-Calibrated
	104 Flow-Calibrated
	105 BasicPositonerValve
	106 AdvancedPositionerValve
	107 DiscreteValve
Usage	Contained
Length	2
Valid View(s)	1, 2, 3, 4
Storage	Non-Volatile

22.15 Trip_Time

Classification	Fieldbus Device
Description	Specifies the amount of time before the alarm enters the tripped condition.
FF Data Type	Float
Range	Positive
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

22.16 Trk_In_D

Classification	Fieldbus Device
Description	This discrete input is used to initiate external tracking of the block output to the value specified by TRL_VAL.
FF Data Type	Value & Status Discrete
Range	N/A
Usage	Input
Length	2
Valid View(s)	1, 3
Storage	Non-Volatile

22.17 Trk_Scale

Classification	Fieldbus Device
Description	The high and low scale values, engineering units code, and number of digits to the right of the decimal point associated with TRK_VAL
FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	4
Storage	Static

22.18Trk_Val

Classification	Fieldbus Device
Description	This input is used as the track value when external tracking is enabled by TRK_IN_D.
FF Data Type	Value & Status Floating Point
Range	N/A
Usage	Input
Length	5
Valid View(s)	1, 3
Storage	Non-Volatile

23. U

23.1 Uncert_Lim

Classification	Fieldbus Device
Description	Defines the limit for Pct_Incl.
FF Data Type	Float
Range	0-100
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

23.2 Unit_Conv

Classification	Fieldbus Device
Description	Specifies the factor to be used to convert the engineering units of Input 1 to engineering units of Input 2.
FF Data Type	Float
Range	N/A
Usage	Contained
Length	4
Valid View(s)	4
Storage	Static

23.3 Update_Evt

Classification	Fieldbus Device
Description	Specifies the event that caused the update.
FF Data Type	Event Update
Range	N/A
Usage	Contained
Length	14
Valid View(s)	N/A
Storage	Dynamic

24. V

24.1 Valve_Man_Id

Classification	Fieldbus Device
Description	The valve manufacturer's identification number as defined by the FOUNDATION Fieldbus.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	4
Storage	Non-Volatile

24.2 Valve_Model_Num

Classification	Fieldbus Device
Description	The valve model number as defined by the valve manufacturer.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	4
Storage	Non-Volatile

24.3 Valve_Sn

Classification	Fieldbus Device
Description	The valve serial number defined by the valve manufacturer..
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	4
Storage	Non-Volatile

24.4 Valve_Type

Classification	Fieldbus Device
Description	The type of valve as defined in FOUNDATION Fieldbus Transducer Specifications (FF-903)..
FF Data Type	Unsigned8
Range	0 Undefined 1 Linear 2 Rotary 255 Other
Usage	Contained
Length	1
Valid View(s)	4b
Storage	Static

24.5 Velocity_Unit

Classification	Fieldbus Device
Description	The units code index for the engineering unit descriptor for the propagation velocity.
FF Data Type	Unsigned16
Range	Refer to the Appendix.
Usage	Contained
Length	2
Valid View(s)	4
Storage	Static

24.6 VendorName

Classification	Fieldbus Device
Description	Defines the vendor or manufacturer name. Resides in The VFD
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

24.7 Vfd

Classification	TPSB Only
Description	Specifies the relative index of the virtual field device in the VFD_REF_ENTRY list.
FF Data Type	Unsigned16
Range	N/A
Usage	N/A
Length	2
Valid View(s)	N/A
Storage	N/A

24.8 VfdTag

Classification	Fieldbus Device
Description	Specifies the tag name for the virtual field device.
FF Data Type	Visible String
Range	N/A
Usage	N/A
Length	32
Valid View(s)	N/A
Storage	N/A

24.9 VIEWS_INDEX

Classification	Fieldbus Device
Description	Specifies the object dictionary index for VIEW_1. Block views are listed contiguously in the object dictionary starting with VIEW_1.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	N/A
Storage	N/A

25. W

25.1 Write_Alm

Classification	Fieldbus Device
Description	This alert is generated if an attempt is made to write to a parameter that has been write-locked.
FF Data Type	Alarm Discrete
Range	N/A
Usage	Contained
Length	13
Valid View(s)	N/A
Storage	Dynamic

25.2 Write_Lock

Classification	Fieldbus Device
Description	If set, no writes from anywhere are allowed, except to clear WRITE_LOCK. Block inputs will continue to be updated.
FF Data Type	Unsigned8
Range	0 Uninitialized 1 Not_Locked 2 Locked
Usage	Contained
Length	1
Valid View(s)	2
Storage	Static

25.3 Write_Pri

Classification	Fieldbus Device
Description	Priority of the alarm generated by clearing the write lock.
FF Data Type	Unsigned8
Range	0-15
Usage	Contained
Length	1
Valid View(s)	4

W
Write_Pri

Storage	Static
---------	--------

26. X

26.1 Xd_Cal_Date

Classification	Fieldbus Device
Description	Displays the date of the last calibration on the transducer block.
FF Data Type	Date
Range	N/A
Usage	Contained
Length	7
Valid View(s)	4c
Storage	Static

26.2 Xd_Cal_Loc

Classification	Fieldbus Device
Description	The physical location at which the last calibration was performed. (ex. NIST, AcmeLabs).
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	4c
Storage	Static

26.3 Xd_Cal_Who

Classification	Fieldbus Device
Description	Specifies the name of the person responsible for the last positioner calibration.
FF Data Type	Visible String
Range	N/A
Usage	Contained
Length	32
Valid View(s)	4c
Storage	Static

26.4 XD_Error

Classification	Fieldbus Device
Description	Transducer error code as defined in the FOUNDATION Fieldbus Transducer Specifications (Block Alarm Subcodes)..
FF Data Type	Unsigned8
Range	0 Other 1 BlockConfiguration 2 LinkConfiguration 3 SimulationActive 4 LocalOverride 5 DeviceFaultState 6 DeviceMaintenance 7 InputFailure 8 OutputFailure 9 MemoryFailure 10 LostStaticData 11 LostNVData 12 ReadbackCheck 13 MaintenanceNeeded 14 PowerUp 15 OutOfService 16 UnspecifiedError 17 GeneralError 18 CalibrationError 19 ConfigurationError 20 ElectronicsFailure 21 MechanicalFailure 22 I/Ofailure 23 DataIntegrityError 24 SoftwareError 25 AlgorithmError
Usage	Contained
Length	1

Valid View(s)	1, 3
Storage	Dynamic

26.5 XD_Scale

Classification	Fieldbus Device
Description	Defines the transducer's high and low scale values, engineering units code, and number of digits to the right of the decimal point used for the value obtained from the transducer.
FF Data Type	Scaling
Range	N/A
Usage	Contained
Length	11
Valid View(s)	2
Storage	Static

26.6 XD_State

Classification	Fieldbus Device
Description	Defines the index to the text describing the states of a discrete transducer.
FF Data Type	Unsigned16
Range	11 Bad::NotConnected:Constant 31 Bad::OutOfService:Constant 67 Uncertain::NonSpecific:Constant 71 Uncertain::LastUsableValue: Constant 75 Uncertain::SubstituteValue: Constant 79 Uncertain::InitialValue: Constant 131 Good_NonCascade::Non Specific:Constant
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

26.7 X_Units

Classification	Fieldbus Device
-----------------------	-----------------

Description	Defines the variable's engineering units x-axis when the variable is to be displayed in a graph.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

27. Y

27.1 Y_Units

Classification	Fieldbus Device
Description	Defines the variable's y-axis engineering units when the variable is to be displayed in a graph.
FF Data Type	Unsigned16
Range	N/A
Usage	Contained
Length	2
Valid View(s)	2
Storage	Static

28. Appendix A —Enumerations

This appendix contains the enumerations of the parameters described in the previous listing of fieldbus parameters.

28.1 ACCEPT_D

Description

Accept_D, PV copy of output switch.

- | | |
|---|--------|
| 0 | Off |
| 1 | Accept |

28.2 ALERTORD

Description

Order of alerts for Alarm_Sum and Ack_Option.

- | | |
|---|-------------|
| 0 | FailAlarm |
| 1 | Reserved |
| 2 | Reserved |
| 3 | reserved |
| 4 | Reserved |
| 5 | Reserved |
| 6 | Reserved |
| 7 | BlockAlarm |
| 8 | AcceptAlarm |
| 9 | IgnoreAlarm |

28.3 ALMSTATE

Description

Alarm State, A discrete enumeration which gives an indication of whether the alert is active and whether it has been activated.

- | | |
|---|---------------------|
| 0 | Uninitialized |
| 1 | Clear Reported |
| 2 | Clear Not_Reported |
| 3 | Active_Reported |
| 4 | Active-Not_Reported |

28.4 ARHTHTYP

Description

Arith_Type, The algorithm types for the Arithmetic Function.

- | | |
|---|------------------------|
| 1 | FlowCompensationLinear |
| 2 | FlowCompensationSQRoot |
| 3 | FlowCompensationAppx |
| 4 | BTUFlow |
| 5 | MultiplyDivide |
| 6 | Average |
| 7 | Summer |
| 8 | ForthOrderPoly |
| 9 | HTGCompensatedLevel |

28.5 AUTH_PDU

Description

FasArTypeAndRole – DLPDU.

- | | |
|---|----------------|
| 0 | Ordinary |
| 1 | Authentication |
| 2 | Source |
| 3 | Maximal |

28.6 CALCERRS

Description

Calc_Errrs, Calculation errors indications for the Calculate function.

- | | |
|---|--------------------------|
| 0 | Non-SpecificError |
| 1 | DivideByZero |
| 2 | Overflow |
| 3 | FunctionError |
| 4 | FunctionCallNotSupported |
| 5 | FuntionCallArgumentCount |
| 6 | FunctionCallArgumentType |
| 7 | IllegalOperator |

8	JMP-OutOfList
9	InvalidDataType
10	TypeConversionFailed
11	ST-Disallowed
12	StackOverflow
13	INST_EX_LIM-Count

28.7 CHEKBACK

Description

CheckBack, The actual status of the final control element in complex AO and DO.

0	Stat_Safeguard_OP
1	Stat_Safeguard_CL
2	Stat_Binary_OP
3	Stat_Binary_CL
4	Stat_Disc
5	Stat_Disc_D_OP
6	Stat_Disc_D_CL
7	Stat_Rb_D_OP
8	Stat_Rb_D_CL
9	Stat_Torque_D_OP
10	Stat_Torque_D_CL
11	Stat_Sub_Mode
12	Stat_Trav_Time
13	Reserved1
14	Reserved2

28.8 COMBTYPE

Description

Comb_Type, Determines how the multiple IN_D values are combined in the Timer function.

0	AND
1	OR
2	ANY2
3	ANY3
21	EXACTLY1
22	EXACTLY2
23	EXACTLY3
40	EVEN
41	ODD

28.9 CONDTYPE

Description

Cond_Type, The type of conditioning applied to PV_D in the Discrete Alarm function.

0	None
1	Delay
2	Extend
3	Debounce
4	Pulse
5	RT_Pulse

28.10 CONSUMER

Description

Pseudo linkage type for the consuming input parameter.

0	NotConf
1	Local
2	Subscriber
3	Client

28.11CURRSUM

Description

Current, The active status of each alarm.

0	Discrete_Alarm
1	HiHi_Alarm
2	Hi_Alarm
3	LoLo_Alarm
4	Lo_Alarm
5	DevHi_Alarm
6	DevLo_Alarm
7	Block_Alarm

28.12DATADELV

Description

FasDLSDAP - DLCEP Data Delivery Feature.

0	Unordered
1	Ordered
2	Disordered
3	Classical

28.13DC_STATE

Description

DC_State, Device Control State represents the current state of the device.

0	ConfirmedPassive
1	ConfirmedState_1
2	ConfirmedState_2
3	GoingToPassive
4	GoingToState_1
5	GoingToState_2
6	FailedPassive
7	FailedState_1

8	FailedState_2
9	Tripped
10	LostInterlock
11	Locked
12	LostMCCPower

28.14DEVCLASS

Description

Device Class – DLOpertatFunctionalClass.

0	UNDEFINED
1	BASIC
2	LINKMASTER
3	BRIDGE

28.15DEVOPTS

Description

Device_Opts, Options used in processing the Device Control function.

0	Interlock
1	TrackSPInLO
2	Permissive
3	ResetRequired
4	Trip
5	LostPower
6	PassiveWhenConfigured
7	BusRestartPassive

28.16DISABLEX

Description

Disable(I), The disable or use input switch for the Input Selector function.

0	Use
---	-----

1 Disable

28.17DISBLSUM

Description

Disabled, The disabled state of each.

0	Disc_Alm_Disabled
1	HiHi_Alm_Disabled
2	Hi_Alm_Disabled
3	LoLo_Alm_Disabled
4	Lo_Alm_Disabled
5	DevHi_Alm_Disabled
6	DevLo_Alm_Disabled
7	Block_Alm_Disabled

28.18DISCCOND

Description

Disc_Cond, Defines the condition in OUT_D that will generate an alarm in the Discrete Alarm function.

0	None
1	FALSE
2	TRUE
3	Change

28.19DLLSDAP

Description

FasDIISDAP - Schedule, Delivery, Authentication, Priority.

0	NotConfig
1	QUU:Acyclic,NA,Source,Urgent
3	QUU:Acyclic,NA,Source,TimeAvail
41	QUB:Acyclic,Disordered,Source,Urgent
42	QUB:Acyclic,Disordered,Source,Norm
43	QUB:Acyclic,Disordered,Source,Time

59	QUB:Acyclic,Clasical,Source,TimeAvai
137	BNU:Cyclic,Unordered,Source,Urgent
153	BNU:Cyclic,Ordered,Source,Urgent

28.20DLLTIMEC

Description

FasDlITimelinessClass - NotConfig or Duplicate Detection.

0	NotConfig
8	BNU:DuplicateDetection

28.21ENBLDSBL

Description

Enable Disable,

0	Uninitialized
1	Disabled
2	Active

28.22FAIL

Description

Fail, Enumeration of the current failure conditions in the Device Control.

1	PassiveConfirmTime
2	State_1ConfirmTime
3	State_2ConfirmTime
4	PassiveConfirmLost
5	State_1ConfirmLost
6	State_2ConfirmLost
7	Tripped
8	LostInterlock
9	LockedOut
10	LostPower
11	MultipleConfirms

28.23FEATSUPR

Description

FmsFeaturesSupported - FMS Services for VCR.

0	GetOD I
1	UnsolicitedStatus
2	PutOD
3	DownLoad
4	UpLoad
5	RequestDomainDownLoad
6	RequestDomainUpload
7	ProgamInvocation
8	StartStopResumeReset
9	Kil
10	Read
11	Write
12	ReadWithType
13	WriteWithType
14	PhysRead
15	PhysWrite
16	InformationReport
17	InformationReportWithType
18	VariableList
19	EventNotification
20	EventNotificationWithType
21	AcknowledgeEventNotification
22	AlterEventConditionMonitoring
23	AddressingWithName
24	GenericInitateDownloadSequence
25	GenericDownloadSegment
26	GenericTerminateDownloadSequence

28.24FMSBYPAS

Description

FasArTypeAndRole , FMS Bypass Switch.

- | | |
|---|----------|
| 0 | NOBYPASS |
| 1 | BYPASS |

28.25FORWREVS

Description

Rev_Flow_n, Reverse flow indications in the Integrator function.

- | | |
|---|---------|
| 0 | Forward |
| 1 | Reverse |

28.26IGNORE

Description

Ignore, Inputs to ignore for the Device Control function.

- | | |
|---|----------------|
| 0 | Permissive |
| 1 | Interlock |
| 2 | Shutdown |
| 3 | PassiveConfirm |
| 4 | State1Confirm |
| 5 | State2Confirm |

28.27INPUTOPT

Description

Input_Opts, Options that control the use of auxiliary inputs in the Arithmetic.

- | | |
|---|--------------------|
| 0 | IN-UseUncertain |
| 1 | IN_LO-UseUncertain |
| 2 | IN_1-UseUncertain |
| 3 | IN_1-UseBad |
| 4 | IN_2-UseUncertain |
| 5 | IN_2-UseBad |
| 6 | IN_3-UseUncertain |

7 IN_3-UseBad

28.28INTEGOPT

Description

Integ_Opts, The configuration of the input type and flow direction in the Totalizer function

- 0 Input1Accumulate
- 1 Input2Accumulate
- 2 FlowForward
- 3 FlowReverse
- 4 UseUncertain
- 5 UseBad
- 6 Carry

28.29INTEGTYP

Description

Integ_Type, Determines the direction of counting and type of resetting in the Integrator function.

- 1 UP_AUTO
- 2 UP_DEM
- 3 DN_AUTO
- 4 DN_DEM
- 5 PERIODIC
- 6 DEMAND
- 7 PER&DEM

28.30INVRTOPT

Description

Invert_Opts, If the bit is set the corresponding input or output is inverted (Timer or Discrete Alarm function).

- 0 InvertIN_D1
- 1 InvertIN_D2
- 2 InvertIN_D3

3	InvertIN_D4
4	InvertOUT_D
5	InvertOut_ALM

28.31MESGTYPE

Description

MessageType - Message Type for Alerts.

0	Undefined
1	EventNotification
2	AlarmClear
3	AlarmOccur

28.32OFFRESET

Description

Reset_D, Reset_In, The off = 0 and reset = 1 state.

0	Off
1	Reset

28.33OPCMDCXO

Description

OP_CMD_CXO, Operator discrete setpoint for complex AO, DO.

0	Reserved
1	Close
2	Open
3	Stop
4	Enable-1-2-3

28.34OPCMDINT

Description

Op_Cmd_Int, Operator Command reset in the Integrator function.

0	Undefined
---	-----------

1 RESET

28.35OPCMDSPG

Description

Op_Cmd_Spg, Operator command to enable the positions in a profile in the Setpoint Ramp function.

0 UNDEFINED
1 RESET_IN
2 ADVANCED
3 REPEAT

28.36OUTALMSM

Description

Out_Alm_Sum, The alarm conditions for a true Out_alm in the Analog Alarm.

0 None
1 LO_LO
2 LO
3 Lows
4 HI
5 Undefined
6 Level1
7 Undefined
8 HI_HI
9 Level2
10 Undefined
11 Undefined
12 Highs
13 Undefined
14 Undefined
15 Any

28.37 PAUSCAUS

Description

Pause_Cause, The items that cause a pause in the Setpoint Ramp function.

0	NotPaused
1	OperatorPause
2	LogicPause
3	DeviationPause
4	Operator&Logic
5	Operator&Deviation
6	Logic&Deviation
7	Operator&Logic&Deviation

28.38 PRIORTY

Description

FasArTypeAndRole - Transport Priority.

1	Urgent
2	Normal
3	TimeAvailable

28.39 QUIESOPT

Description

Quies_Opt, Behavior options for Out while in quiescence in the timer function.

0	Uninitialized
1	Clear
2	Last

28.40 REASON

Description

Reason_Code, The reasons why a Function Block service fails.

1	ParameterCheck
---	----------------

Description

Reason_Code, The reasons why a Function Block service fails.

2	ExceedsParameterLimits
5	DataValueNeverWritable
7	DeleteUnsucessfull
3	WrongModeForRequest
4	WriteProhibitedByLock
6	InstantiateUnsucessfull
8	AlarmAlreadyAcked
9	Undefined
10	Undefined
11	Undefined
12	Undefined
13	Undefined
14	Undefined
15	Undefined
16	WarningDataValueSet
17	WarningSetToNearestValue
18	ParameterNotImplemented
19	IncorrectFormat
20	InvalidDate
21	ModeError
23	AutomaticZeroError
22	ProcessOutOfRange
24	DeviceBusy
25	UpdateFailed
26	InvalidConfigurationForRequest
27	ExcessCorrection
28	CalibrationFailed
29	CalibrationInProgress

28.41 RESACTSP

Description

FasDIIResidualActivitySupported.

0 FALSE

255 TRUE

28.42 SAMPTYPE

Description

Sample Type - Trend Sample Type.

0 NotInitialized

1 SampleValue

2 AverageValue

28.43 SCHEDLAS

Description

FasDLSDAP - Schedule LAS Acyclic, Cyclic.

0 Acyclic

1 Cyclic

28.44 SELECT_T

Description

Select_Type, selector action types.

0 Uninitialized

1 FirstGood

2 Min

3 Max

4 Mid

5 Avg

28.45 SERVICE

Description

Service Operation, Service Type for Link Objects.

0 Undefined

- | | |
|---|-------------|
| 1 | Local |
| 2 | Publisher |
| 3 | Subscriber |
| 4 | Client |
| 5 | Server |
| 6 | AlertNotify |
| 7 | TrendReport |

28.46 SMSUPPORT

Description

SM (System Management) Supported Features.

- | | |
|----|-----------------|
| 0 | SET_PDTAG_AGENT |
| 1 | SET_ADDR_AGENT |
| 2 | CLR_PDTAG_AGENT |
| 3 | IDENTIFY_AGENT |
| 4 | LOC_FB_AGENT |
| 5 | SET_PDTAG_MGR |
| 6 | SET_ADDR_MGR |
| 7 | CLR_ADDR_MGR |
| 8 | IDENTIFY_MGR |
| 9 | LOC_FB_MGR |
| 10 | FMS_SERVER |
| 11 | TIME_SLAVE |
| 12 | SCHEDULE_FB |
| 13 | TIME_PLUSHER |

28.47 SPGSTATE

Description

Spg_State, Operation states of the block in the Setpoint Ramp function.

- | | |
|---|-----------|
| 0 | Undefined |
| 1 | Ready |
| 2 | Active |

- | | |
|---|--------|
| 3 | Pause |
| 4 | AT_End |

28.48STANDTYP

Description

StandardTypes - Standard Type of Alerts.

- | | |
|----|-------------|
| 0 | Undefined |
| 1 | Lo |
| 2 | HI |
| 3 | LoLo |
| 4 | HiHi |
| 5 | DvLo |
| 6 | DvHi |
| 7 | Disc |
| 8 | Block |
| 9 | UpdateData |
| 10 | Write |
| 11 | UpdateLink |
| 12 | UpdateTrend |

28.49STRTTYPE

Description

Start_Type, The starting point options in the Setpoint Ramp function.

- | | |
|---|---------------|
| 0 | Uninitialized |
| 1 | UseCurve |
| 2 | UseDuration |
| 3 | UseRate |

28.50TIMECLAS

Description

DL Timeliness Class.

0	None
1	Resident
2	Update
3	Synchronous
4	Transparent
5	Reserve1
6	Reserve2
7	Reserve3

28.51 TIMEUNIT

Description

Time_Units, The selector for time units for display in the Pulse Input, Setpoint ramp and Timer functions.

0	Uninitialized
1	Seconds
2	Hours
3	Minutes
4	Days
5	[day-[hr:[min[:sec]]]]

28.52 TIMRTYPE

Description

Timer_Type, The type of time processing applied to PV in the Timer function.

0	Measure
1	Accum
2	Compare
3	Delay
4	Extend
5	Debounce
6	Pulse
7	RT_Pulse

28.53TRUFALSE

Description

Boolean, The state of False = 0, True = 1.

0	FALSE
1	TRUE

28.54TYPEROLE

Description

VCR Type and Role together.

0	NotConfig
50	Server,QUB
68	Source,QUU
84	Sink,QUU
102	Publisher,BNU
118	Subscriber,BNU
162	Client,QUB

28.55UNACKNOW

Description

UNACKNOWLEDGED, A discrete enumeration which is set to Unacknowledged when an alarm occurs, and set to Acknowledged by a write from a human interface device or other entity which can acknowledge that the alarm/event has been noticed.

0	Undefined
1	Acknowledged
2	Unacknowledged

28.56UNACKSUM

Description

Unacknowledge, The unacknowledged state of each alarm.

0	Disc_Alm_Unack
1	HiHi_Alm_Unack

2	Hi_Alm_Unack
3	LoLo_Alm_Unack
4	Lo_Alm_Unack
5	DevHi_Alm_Unack
6	DevLo_Alm_Unack
7	Block_Alm_Unack

28.57UNITS

Description

The engineering units associated with a PV

1000	K (Kelvin)
1001	°C (Celsius)
1002	°F (Fahrenheit)
1003	°R (Rankine)
1004	r (radian)
1005	° (degree)
1006	' (minute)
1009	rev (revolution)
1010	m (meter)
1011	km (kilometer)
1012	cm (centimeter)
1013	mm (millimeter)
1014	μm (micrometer)
1032	yd ²
1033	mile ²
1034	m ³ (cubic meter)
1040	ml
1041	hl (milliliter)
1043	ft ³
1044	yd ³ (cubic yard)
1045	mile ³ (cubic mile)
1046	pint
1047	quart

Description

The engineering units associated with a PV

1048	gallon
1049	ImpGal
1052	bbl (liq)
1053	SCF
1054	s
1055	ks
1056	ms (millisecond)
1057	μs (microsecond)
1060	d
1061	m/s
1067	ft/s
1068	yd/s
1070	ft/min
1071	yd/min
1072	in/h
1073	ft/h
1074	yd/h
1075	MPH
1076	m/s ² (meter per second per second)
1077	Hz
1078	THz
1079	GHz
1080	MHz
1081	kHz
1082	1/s
1083	1/min
1084	rev/s (revolutions per second)
1085	RPM
1086	r/s (radian per second)
1087	1/s ²
1088	kg
1089	g

Description

The engineering units associated with a PV

1090	mg (milligram)
1091	Mg (megagram)
1092	t (metric ton)
1093	oz
1094	lb
1095	Ston (short ton = 2000 pounds)
1096	Lton (long ton = 2240 pounds)
1097	kg/m ³ (kilograms per cubic meter)
1098	Mg/m ³ (megagrams per cubic meter)
1099	kg/dm ³ (kilograms per decimeter)
1100	g/cm ³ (grams per cubic centimeter)
1101	g/m ³ (grams per cubic meter)
1103	kg/L (kilograms per liter)
1104	g/ml (grams per milliliter)
1105	g/L (grams per liter)
1106	lb/in ³ (pounds per cubic inch)
1107	lb/ft ³ (pounds per cubic foot)
1108	lb/gal (pound per gallon)
1110	DegTwad (degrees Twaddell = 200SGU-200)
1111	DegBaum hv (degrees Baume heavy = 0 for 1.842 SGU)
1118	kg-m ² (kilogram square meter)
1120	N (newton; 1N=1kg-m/s ²)
1121	MN (meganewton)
1122	kN (kilonewton)
1123	mN (millinewton)
1124	μN (micronewton))
1125	kg-m ² /s (kilogram square meter per second)
1126	N-m (newton meter)
1127	MN-m (meganewton meter)
1128	kN-m (kilonewton meter)

Description

The engineering units associated with a PV

1129	mN-m (millinewton meter)
1130	Pa (pascal)
1131	GPa (gigapascal)
1132	MPa (megapascal)
1133	kPa (kilopascal)
1134	mPa (millipascal)
1135	μPa (micropascal)
1136	hPa (hectopascal)
1137	bar (1 bar = 100 kPa)
1138	mbar (millibar; 1 millibar = 1 hPa)
1139	torr (1 torr = 1mmHg)
1140	atm (atmospheres = 14.7 psia; 760 mmHg)
1141	psi (pounds per square inch)
1142	psia (pounds per square inch absolute)
1143	psig (pounds per square inch guage)
1144	g/cm ² (grams per square centimeter)
1145	kg/cm ² (kilograms per sqare centimeter)
1146	inH2O (inches of water)
1147	inH2O (4°C)
1148	inH2O (68°F)
1149	mmH2O (millimeters of water; 1 mmH2O = .098 hPa)
1150	mmH2O (4°C) (millimeters of water at 4°C)
1151	mmH2O (68°F) (millimeters of water at 68°F)
1152	ftH2O (feet of water)
1153	ftH2O (4°C) (feet of water at 4°C)
1154	ftH2O (68°F) (feet of water at 68°F)
1155	inHg (inches of mercury; 1 inHg = 13.6 inH2O)
1156	inHg (0°C) (inches of mercury at 0° C)
1157	mmHg (millimeters of mercury; 1 mmHg = 1.333 hPa))

Description

The engineering units associated with a PV

1158	mmHg (0°C) (millimeters of mercury at 0°C)
1159	Pa-s (Pascal second)
1160	M ² /s (square meter per second)
1161	P (poise)
1162	cP (centipoise; 1cP=1 mPa-s absolute viscosity)
1163	St (stokes)
1164	cSt (centistokes; 1cSt=1mm ² /s kinematic viscosity)
1165	N/m (newton per meter)
1166	mN/m (millinewton per meter)
1167	J (joules; 1 J=1N-m)
1168	EJ (exajoules))
1169	PJ (petajoules)
1170	TJ (terajoules)
1171	GJ (gigajoules)
1172	MJ (megajoules)
1173	kJ (kilojoules)
1174	mJ (jillijoules)
1175	WH (watt hour; 1 W-h=3.6 kJ)
1176	TWH (terawatt hour)
1177	GWH (gigawatt hour)
1178	MWH (megawatt hour)
1179	KWH (kilowatt hour)
1180	cal (calorie = 4.184J)
1181	kcal (kilocalorie)
1182	Mcal (metacalorie)
1183	Btu (British thermal unit; 1 Btu=0.2519958 kcal)
1187	TW (terawatt)
1188	GW (gigawatt)
1189	MW (megawatt)
1190	KW (kilowatt)

Description

The engineering units associated with a PV

1191	mW (milliwatt)
1192	μ W (microwatt)
1193	nW (nanowatt)
1194	pW (picowatt)
1195	Mcal/h (megacalorie per hour)
1196	MJ/h (megajoule per hour)
1197	Btu/h (British thermal unit per hour)
1198	Hp (horse power; = 746 W)
1199	W/(m-K) (watts per meter kelvin)
1200	W/(m ² -K) (watts per square meter kelvin)
1201	m ² -K/W (square meter kevin per watt)
1202	J/K (joules per kelvin)
1203	kJ/K (kilojoule per kelvin)
1204	J/(kg-K) (joule per kilogram kelvin)
1205	kJ/(kg-K) (kilojoule per kilogram kelvin)
1206	J/kg (joule per kilogram)
1207	MJ/kg (megajoule per kilogram)
1208	kJ/kg (kilojoule per kilogram)
1209	A (ampere)
1210	kA (kiloampere)
1211	mA (milliampere)
1212	μ A (microampere)
1213	nA (nanoampere)
1214	pA (picoampere)
1215	C (coulomb)
1216	MC (megacoulomb)
1217	kC (kilocoulomb)
1218	μ C (microcoulomb)
1219	nC (nanocoulomb)
1220	pC (picocoulomb)
1221	A-h (ampere hour) (1A-h = 3.6kC)
1222	C/m ³ (coulomb per cubic meter)

Description

The engineering units associated with a PV

1223	C/mm ³ (coulomb per cubic millimeter)
1224	C/cm ³ (coulomb per cubic centimeter)
1225	kC/m ³ (kilocoulomb per cubic meter)
1226	mC/m ³ (millicoulomb per cubic meter)
1227	μC/m ³ (microcoulomb per cubic meter)
1228	C/m ²
1229	C/mm ²
1230	C/cm ²
1231	kC/m ²
1232	mC/m ²
1233	mC/m ²
1234	V/m (volt per meter)
1235	MV/m (megavolt per meter)
1236	kV/m (kilovolt per meter)
1237	V/cm
1238	mV/m (millivolt per meter)
1239	μV/m (microvolt per meter)
1240	V (volt = 1 W/A)
1241	MV (megavolt)
1242	KV (kilovolt)
1243	mV (<i>millivolt</i>)
1244	μV (<i>microvolt</i>)
1245	F (<i>farad</i> = 1 C/V)
1246	mF
1247	μF
1248	nF
1249	pF
1250	F/m
1251	μF/m
1252	nF/m
1253	pF/m
1254	C-m (<i>coulomb meter</i>)

Description

The engineering units associated with a PV

1255	A/m ²
1256	MA/m ²
1257	A/cm ²
1258	kA/m ²
1259	A/m
1260	kA/m
1261	A/cm
1262	T (<i>tesla</i> = 1 Wb/m ²)
1263	mT (<i>millitesla</i>)
1264	μT
1265	nT
1266	Wb (<i>weber</i> = 1 V-s)
1267	mWb
1268	Wb/m
1269	kWb/m
1270	H (<i>henry</i> = 1 Wb/A)
1271	mH
1272	μH
1273	nH
1274	pH
1275	H/m
1276	μH/m
1277	nH/m
1278	A-m ²
1279	N-m ² /A (<i>newton square meter per ampere</i>)
1280	Wb-m (<i>weber meter</i>)
1281	Ohm (<i>Ohm</i> ; 1 ohm.Ω = 1 V/A)
1282	GOhm (<i>gigaOhm</i>)
1283	MOhm (<i>megaOhm</i>)
1284	kOhm (<i>kiloOhm</i>)
1285	mOhm (<i>milliOhm</i>)
1286	μOhm (<i>microOhm</i>)

Description

The engineering units associated with a PV

1287	S (<i>siemens; 1 S = 1 /ohmΩ^{-2}</i>)
1288	kS
1289	mS
1290	μ S
1291	Ohm-m (<i>ohm meter</i>)
1292	GOhm-m
1293	MOhm-m
1294	kOhm-m
1295	Ohm-cm
1296	mOhm-m
1297	μ Ohm-m
1298	nOhm-m
1299	S/m (<i>siemens per meter</i>)
1300	MS/m
1301	kS/m
1302	mS/cm
1303	μ S/mm
1304	1/H (<i>per henry</i>)
1305	sr (<i>steradian</i>)
1306	W/sr (<i>watt per steradian</i>)
1307	W/(sr-m ²)
1308	W/(m ²)
1309	lm (<i>lumen; 1 lumen = 1 cd-sr</i>)
1310	lm-s (<i>lumen second</i>)
1311	lm-h (<i>lumen hour = 3600 lm-s</i>)
1312	lm/m ²
1313	lm/W
1314	lx (<i>lux = 1 lm/m²</i>)
1315	lx-s (<i>lux second</i>)
1316	cd (<i>candela; SI</i>)
1317	cd/m ²
1318	g/s (<i>gram per second</i>)

Description

The engineering units associated with a PV

1319	g/min (<i>gram per minute</i>)
1320	g/h (<i>gram per hour</i>)
1321	g/d (<i>gram per day</i>)
1322	kg/s (<i>gram per second</i>)
1323	kg/min (<i>gram per minute</i>)
1324	kg/h (<i>gram per hour</i>)
1325	kg/d (<i>gram per day</i>)
1326	t/s (<i>metric ton per second; 1 t = 10³kg</i>)
1327	t/min
1328	t/h
1329	t/d
1330	lb/s
1331	lb/min
1332	lb/h
1333	lb/d
1334	STon/s (<i>short ton per second</i>)
1335	STon/min
1336	STon/h
1337	STon/d
1338	LTon/s (<i>long ton per second</i>)
1339	LTon/min
1340	LTon/h
1341	LTon/d
1342	%
1343	% sol/wt (<i>percent solids per weight</i>)
1344	% sol/vol (<i>percent solids per volume</i>)
1345	% stm qual (<i>percent solids per quality</i>)
1346	% plato (<i>percent plato; % sugar by weight</i>)
1347	m ³ /s (cubic meters per second)
1348	m ³ /min (cubic meters per minute)
1349	m ³ /h (cubic meters per hour)
1350	m ³ /d (cubic meters per day)

Description

The engineering units associated with a PV

1351	L/s (<i>liter per second</i>)
1352	L/min (<i>liter per minute</i>))
1353	L/h (<i>liter per hour</i>)
1354	L/d (<i>liter per day</i>)
1355	ML/d (<i>megaliter per day</i>)
1356	CFS (<i>cubic feet per second</i>)
1357	CFM (<i>cubic feet per minute</i>)
1358	CFH (<i>cubic feet per hour</i>)
1359	ft ³ /d (<i>cubic feet per day</i>)
1360	SCFM (<i>cubic feet per second</i>)
1361	SCFH (<i>cubic feet per hour</i>)
1362	gal/s (<i>US gallons per second</i>)
1363	GPM (<i>US gallons per minute</i>)
1364	gal/h (<i>US gallons per hour</i>)
1365	gal/d (<i>US gallons per day</i>)
1366	Mgal/d (<i>mega US gallon per day</i>)
1367	ImpGal/s (<i>Imperial gallon per second</i>)
1368	ImpGal/min (<i>Imperial gallon per minute</i>)
1369	ImpGal/h (<i>Imperial gallon per hour</i>)
1370	ImpGal/d (<i>Imperial gallon per day;</i> <i>1bbl=42US gallons</i>)
1371	bbl/s (<i>barrel per second</i>)
1372	bbl/min (<i>barrel per minute</i>)
1373	bbl/h (<i>barrel per hour</i>)
1374	bbl/d (<i>barrel per day</i>)
1375	W/m ² (<i>watt per square meter</i>)
1376	mW/m ² (<i>milliwatt watt per square meter</i>)
1377	μW/m ² (<i>microwatt per square meter</i>)
1378	pW/m ² (<i>picowatt per square meter</i>)
1379	Pa-s/m ³ (<i>pascal second per cubic meter</i>)
1380	N-s/m (<i>newton second per meter</i>)
1381	Pa-s/m (<i>pascal second per meter</i>)

Description

The engineering units associated with a PV

1382	B (<i>bel</i>)
1383	dB (<i>decibel</i> = 10^{-1} B)
1384	mol (<i>mole</i>)
1385	kmol (<i>kilomole</i>)
1386	mmol (<i>millimole</i>)
1387	μmol (<i>micromole</i>)
1388	kg/mol (<i>kilogram per mole</i>)
1389	g/mol (<i>gram per mole</i>)
1390	m ³ /mol (<i>cubic meter per mole</i>)
1391	dm ³ /mol (<i>cubic decimeter per mole</i>)
1392	cm ³ /mol (<i>cubic centimeter per mole</i>)
1393	L/mol (<i>liters per mole</i>)
1394	J/mol (<i>joule per mole</i>)
1395	kJ/mol (<i>kilojoule per mole</i>)
1396	J/(mol·K) (<i>joule per mole kelvin</i>)
1397	mol/m ³ (<i>mole per cubic meter</i>)
1398	mol/dm ³ (<i>mole per cubic decimeter</i>)
1399	mol/L (<i>mole per liter</i>)
1400	mol/kg (<i>mole per kilogram</i>)
1401	mmol/kg (<i>millimole per kilogram</i>)
1402	Bq (<i>becquerel</i> ; 1 Bq = 1·s ⁻¹)
1403	MBq (<i>megabecquerel</i>)
1404	kBq (<i>kilobecquerel</i>)
1405	Bq/kg (<i>becquerel per kilogram</i>)
1406	kBq/kg (<i>kilobecquerel per kilogram</i>)
1407	MBq/kg (<i>megabecquerel per kilogram</i>)
1408	Gy (<i>gray</i> = 1 J/kg)
1409	mGy (<i>milligray</i>)
1410	rad (<i>rad</i> = 10^{-2} Gy)
1411	Sv (<i>sievert</i> ; 1 Sv = 1 J/kg)
1412	mSv (<i>millisievert</i>)
1413	rem (<i>rem</i> ; 1 rem = 10^{-2} Sv)

Description

The engineering units associated with a PV

1414	C/kg (<i>coulomb per kilogram</i>)
1415	mC/kg (<i>millicoulomb per kilogram</i>)
1416	R (<i>röntgen</i> ; $1\text{ R} = 2.58 \times 10^{-4}\text{ C/kg}$)
1417	1/J-m ³
1418	e/V-m ³
1419	m ³ /C (<i>cubic meter per coulomb</i>)
1420	V/K (<i>volt per kelvin</i>)
1421	mV/K
1422	pH
1423	ppm
1424	ppb
1425	ppt
1426	degBrix (<i>degrees Brix</i> ; % sugar by weight)
1427	degBall (<i>degrees Balling</i> ; % sugar by weight)
1428	proof/vol
1429	proof/mass
1430	lb/ImpGal
1431	kcal/s (<i>kilocalorie per second</i>)
1432	kcal/min
1433	kcal/h
1434	kcal/d
1435	Mcal/s (<i>megacalorie per second</i>)
1436	Mcal/min
1437	Mcal/d
1438	kJ/s (<i>kilojoules per second</i>)
1439	kJ/min
1440	kJ/h
1441	kJ/d
1442	MJ/s (<i>megajoules per second</i>)
1443	MJ/min
1444	MJ/d

Description

The engineering units associated with a PV

1445	Btu/s
1446	Btu/min
1447	Btu/day
1448	μgal/s (<i>micro US gallon per second</i>)
1449	mgal/s (<i>milli US gallon per second</i>)
1450	kgal/s
1451	Mgal/s
1452	μgal/min
1453	mgal/mi
1454	kgal/min
1455	Mgal/min
1456	μgal/h
1457	mgal/h
1458	kgal/h
1459	Mgal/h
1460	μgal/d
1461	mgal/d
1462	kgal/d
1463	μImpGal/s (<i>micro imperial gallon per second</i>)
1464	mImpGal/s
1465	kImpGal/s
1466	MImpGal/s
1467	μImpGal/min
1468	mImpGal/min
1469	kImpGal/min
1470	MImpGal/min
1471	μImpGal/h
1472	mImpGal/h
1473	kImpGal/h
1474	MImpGal/h
1475	μImpGal/d

Description

The engineering units associated with a PV

1476	mImpGal/d
1477	kImpGal/d
1478	MImpGal/d
1479	μbbl/s (<i>microbarrel per second</i>)
1480	mbbl/s (<i>millibarrel per second</i>)
1481	kbbbl/s (<i>kilobarrel per second</i>)
1482	Mbbl/s (<i>megabarrel per second</i>)
1483	μbbl/min
1484	mbbl/min
1485	kbbbl/min
1486	Mbbl/min
1487	μbbl/h
1488	mbbl/h
1489	kbbbl/h
1490	Mbbl/h
1491	μbbl/d
1492	mbbl/d
1493	kbbbl/d
1494	Mbbl/d
1495	μm ³ /s (<i>cubic micrometer per second</i>)
1496	mm ³ /s
1497	km ³ /s
1498	Mm ³ /s
1499	μm ³ /min
1500	mm ³ /min
1501	km ³ /min
1502	Mm ³ /min
1503	μm ³ /h
1504	mm ³ /h
1505	km ³ /h
1506	Mm ³ /h
1507	μm ³ /d

Description

The engineering units associated with a PV

1508	mm ³ /d
1509	km ³ /d
1510	Mm ³ /d
1511	cm ³ /s (<i>cubic centimeter per second</i>)
1512	cm ³ /min
1513	cm ³ /h
1514	cm ³ /d
1515	kcal/kg (<i>kilocalorie per kilogram</i>)
1516	Btu/lb
1517	kL (<i>kiloliter</i>)
1518	kL/min
1519	kL/h
1520	kL/d
1521	unit_code1521
1522	unit_code1522
1523	unit_code1523
1524	unit_code1524
1525	unit_code1525
1526	unit_code1526
1527	unit_code1527
1528	unit_code1528
1529	unit_code1529
1530	unit_code1530
1531	unit_code1531
1532	unit_code1532
1533	unit_code1533
1534	unit_code1534
1535	unit_code1535
1536	unit_code1536
1537	unit_code1537
1538	unit_code1538
1539	unit_code1539

Description

The engineering units associated with a PV

1540	unit_code1540
1541	unit_code1541
1542	unit_code1542
1543	unit_code1543
1544	unit_code1544
1545	unit_code1545
1546	unit_code1546
1547	unit_code1547
1548	unit_code1548
1549	unit_code1549
1550	unit_code1550

28.58UNREPSUM

Description

Unreported, The unreported status of each Alarm.

0	Disc_Alm_Unrep
1	HiHi_Alm_Unrep
2	Hi_Alm_Unrep
3	LoLo_Alm_Unrep
4	Lo_Alm_Unrep
5	DevHi_Alm_Unrep
6	DevLo_Alm_Unrep
7	Block_Alm_Unrep

28.59UPDSTATE

Description

Update State, A discrete enumeration which gives an indication of whether the alert has been reported.

0	Undefined
1	Reported
2	Not_Reported

28.60 VCRROLE

Description

VCR send-receive Roles

0	NotConfig
1	Responder
2	Undefined
3	Server
4	Source
5	Sink
6	Publisher
7	Subscriber
8	Undefined
9	Initiator
10	Client

28.61 VCRTYPE

Description

VCR Data Transport Type

0	NotConfig
1	QUB
2	QUU
3	BNU

Reader Comments

Fax Transmittal		Fax No.: (602) 313-4212	
Reader Comments			
To: Information Development			
From:	Name	Date:	
	Title:		
	Company:		
	Address:		
	City:	State:	Zip:
	Telephone:	Fax:	
Parameter Reference Dictionary, FB09-611A			
Release R611, 02/01			
Comments:			
You may also call 800-343-0228 (available in the 48 contiguous states), or write to:			
Honeywell		Fieldbus	
Industrial Automation and Control			
16404 North Black Canyon Highway		Knowledge Building Tools	
Phoenix, AZ 85053-3033			

Honeywell

Honeywell
Industrial Automation and Control
16404 North Black Canyon Highway
Phoenix, AZ 85053